MACHINE LEARNING MODELS FOR DETECTION OF URL-BASED PHISHING ATTACKS

A RESEARCH PROJECT

BY

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SUBMITTED TO

DEPARTMENT OF COMPUTER SCIENCE,
FACULTY OF COMPUTING,
UNIVERSITY OF UYO, UYO.

IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF BACHELOR OF SCIENCE(B.Sc) IN COMPUTER SCIENCE.

SEPTEMBER, 2025

DECLARATION

I affirm that this project titled **"MACHINE LEARNING MODELS FOR DETECTION OF URL-BASED PHISHING ATTACKS"** is my original research work, and I take full responsibility for its content. The information presented in this project is supported by specific references, acknowledging all sources utilized.

……..……

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CERTIFICATION

This is to certify that this project on **"MACHINE LEARNING MODELS FOR DETECTION OF URL-BASED PHISHING ATTACKS"** was written by **OKON, INEMESIT UDO** with registration number **20/SC/CO/890** under my supervision

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DEDICATION

I dedicate this work to God almighty, who granted me the wisdom and grace to make this work a reality and also to my Big Uncle, Mr Sunday James Akpan, My Foster Mother and the Children: Mrs Comfort Romanus, Mrs Elizabeth Romanus, Mrs Blessing Daniel whose encouragement and sacrifices strengthened me throughout my studies at the University of Uyo,Uyo, Akwa Ibom

ACKNOWLEDGEMENT

I am deeply grateful to the Almighty God for His wisdom, strength, and guidance throughout this research work. His grace has been my anchor, making this project a reality.

I extend my sincere appreciation to my project supervisor, **Dr. Daniel E. Asuquo**, for his invaluable guidance, constructive criticism, and unwavering support during the course of this research. His expertise and encouragement were instrumental in shaping this work.

My heartfelt thanks go to **Dr. (Mrs.) Patience Usoro Usip**, the Head of the Department of Computer Science, University of Uyo, for providing an enabling environment for learning and research.

I also express my thanks to my amiable and ever supporting lecturers, Prof. (Mrs) Uduak A. Umoh (Also the pioneer Dean Faculty of Computing), Prof. Inyang G. Udoinyang(also the Director Centre of Excellence on Computational Intelligence), Prof. Okure U, Obot, Prof. Moses Ekpenyong, Prof. Victor Ekong (the pioneer HOD Software Engineering), Prof.(Mrs) Uyinomem Ekong (the pioneer HOD, Cyber security) , Dr. Samuel S. Udoh (the Pioneer HOD Data Science and also Staff Adviser Department of Computer Science), Dr. Daniel E. Asuquo(Pioneer HOD, Information Technology and also Staff Adviser Faculty of computing) Dr.(Mrs) Patience Usoro Usip, Dr. U.D. George, Dr. Ifiok J. Udoh, Dr. Edward N. Udoh, Dr.(Mrs) Imo J. Eyoh, Dr. I.J. Umoeka, Dr. Samuel A. Robinson, Mr S. O. Adeoye, Mr D. Ambrose

I am also indebted to my parents and family for their love, prayers, and sacrifices, which kept me motivated throughout this journey. Their unwavering belief in me has been a constant source of inspiration.

I acknowledge the contributions of my colleagues and friends especially my earnest university paddies Unyime Udongwo, Iniubong Usoro (Sazi), Ekemini Udongwo (Kemzzy), Blessed Omerenma, Mercy Maximum, Esther Ubak and Daniel Effiong (SigmaDev), who provided moral support, shared ideas, and offered assistance during challenging moments.

Finally, I appreciate the authors and researchers whose works were referenced in this study. Their contributions to the field of cybersecurity and machine learning provided a solid foundation for this project.

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ABSTRACT

Phishing attacks remain a major cybersecurity threat, using fake websites to trick users into revealing sensitive information like passwords and credit card details. Current detection methods such as blacklists of known malicious sites, struggles to keep up with new phishing tactics. This research tackles this problem by developing a lightweight, real-time machine learning models for detecting phishing URLs using Logistic Regression and Multinomial Naive Bayes algorithms. A large dataset of 549,346 entries of legitimate and phishing URLs was collected from Kaggle, followed by preprocessing steps including duplicate removal, tokenization with RegexpTokenizer, stemming using SnowballStemmer, and conversion into a numerical feature matrix using CountVectorizer. These processes generated a high-dimensional sparse matrix consisting of thousands of URL tokens representing lexical, host-based, and behavioral features. The machine learning models were implemented in a Python cloud-based environment (Google Colab) and trained using two features URLs and Label. Result shows that LR performs slightly better (96.5% accuracy), than MNB (95.7%), making it more reliable for real world used. Both machine learning models were tested on Accuracy, Precision, Recall and F1-score. The Phishing detection models was designed to be performed on real time and also efficiently light weight, meaning it can run effectively even on devices with limited processing power.

CHAPTER ONE

INTRODUCTION

1.1 Background of Study

Phishing attack is a prevalent cyber threat, involving deceptive tactics where attackers impersonate legitimate entities to steal sensitive information such as login credentials and financial data (Ibrokhimov et al., 2024). These attacks frequently utilize fraudulent Uniform Resource Locators (URLs) designed to resemble trusted websites, exploiting human trust and technical vulnerabilities. The growing complexity and frequency of phishing campaigns have made them a critical security challenge, particularly in developing economies where cybersecurity awareness is still evolving (Musa et al., 2023). These attacks lead to substantial financial losses, identity theft, and organizational reputational harm (Wang et al.,2024).

Traditional methods of phishing detection, such as blacklist-based approaches, are often ineffective against rapidly evolving phishing techniques, especially in regions with high mobile penetration rates (Balogun et al., 2024). Machine learning (ML) offers a promising solution by leveraging pattern recognition and predictive modeling to identify malicious URLs based on their features (Zhang et al., 2023). ML algorithms can analyze various URL characteristics, including lexical features, host-based features, and potentially traffic-based features, to detect phishing attempts with high accuracy (Adewumi et al., 2023).

Figure 1.1 illustrates how an attacker can lure a victim to access a phishing website .

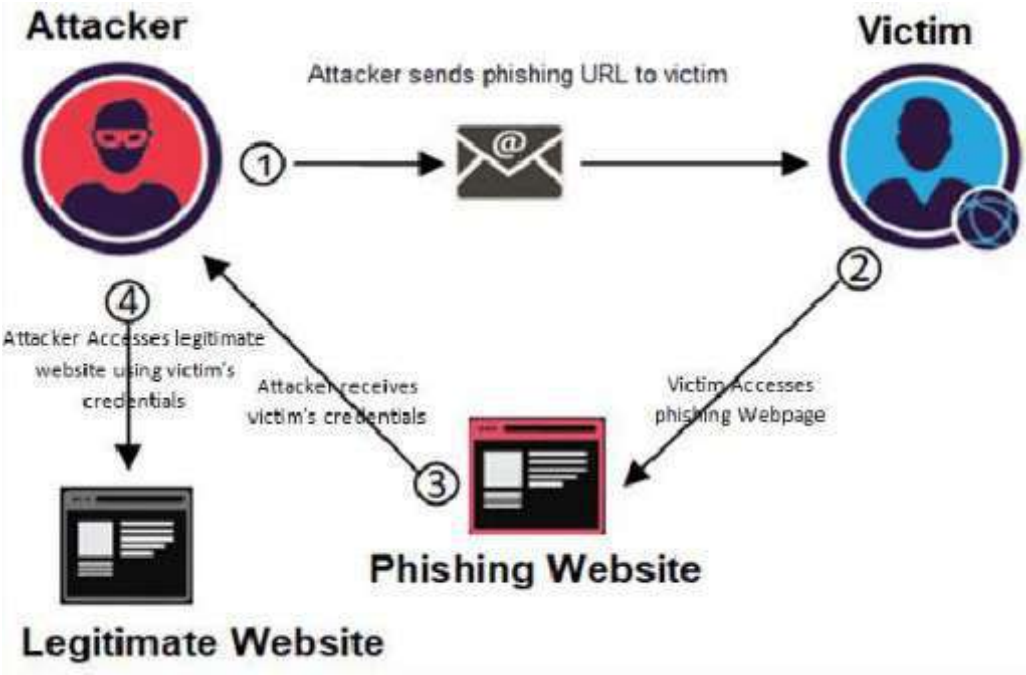


Figure 1.1: Illustration of Phishing Attack

1.2 Statement of Problem

Phishing attacks continue to evolve, using deceptive URLs to steal sensitive information, yet many existing detection systems struggle with low accuracy, high false positives, and inability to adapt to new attack patterns.

Traditional blacklist-based methods are reactive and fail to detect zero-day phishing URLs, while some machine learning solutions are too resource-intensive for real-time use. This project addresses these gaps by developing a lightweight, real-time phishing detection models using Logistic Regression and Multinomial Naive Bayes to improve detection rates while minimizing computational overhead.

1.3 Aim and Objectives

The aim of this project is to develop a lightweight and real-time machine learning-based models for detecting phishing URLs using Logistic Regression and Multinomial Naive Bayes.

To achieve the aim, the following objectives have been set:

i. To collect and preprocess a comprehensive dataset of legitimate and phishing URLs from kaggle.
ii. To extract lightweight and relevant features from the URLs.
iii. To design and implement a framework incorporating machine learning Logistic Regression and Multinomial Naive Bayes for light weight URL phishing detection.
iv. To evaluate the performance of these models using appropriate metrics.
v. To compare the two machine learning models and identify the most suitable one for detecting URL phishing.

1.4 Methodology

This project is aim to develop a machine learning models for detection of Uniform Resource Locators URL-based phishing attacks, The methodology includes:

1.4.1 Dataset Collection

The dataset used in this study was obtained from Kaggle, containing both legitimate (good) and phishing (bad) URLs. It consists of 549,347entries, with each record having a URL and a

corresponding Label indicating whether it is phishing or legitimate. The dataset was loaded using the Pandas library in Python for further preprocessing.

1.4.2 Data Preprocessing

The phishing URL dataset from Kaggle was cleaned by removing irrelevant characters, tokenizing URLs using RegexpTokenizer, and normalizing text through SnowballStemmer for stemming. The processed tokens were finally transformed into a numerical feature matrix using CountVectorizer (Bag-of-Words), producing structured input for the Logistic Regression and Multinomial Naive Bayes models.

1.4.3 Model Selection

Two supervised learning algorithms were selected for the phishing detection: Logistic Regression and Multinomial Naive Bayes. These models were chosen for their effectiveness in text classification tasks, computational efficiency, light weight and ability to handle high-dimensional sparse feature matrices generated from the CountVectorizer.

1.4.4 Model Implementation

The Two machine learning algorithms Logistic Regression and Multinomial Naive Bayes were implemented using (Jupyter notebook Python cloud-based environment) google Colab.

1.4.5 Model Training and Optimization

The models were trained using 80% of the dataset, with the remaining 20% reserved for testing. To improve performance, Grid Search was used for hyperparameter tuning, and cross-validation was applied to prevent overfitting. Due to the imbalance in the dataset (fewer phishing URLs than legitimate ones), undersampling using RandomUnderSampler was applied to reduce the majority class, creating a balanced training set. This method was chosen over Synthetic Minority Oversampling Technique (SMOTE) because the data size was large, the process helps to reduces the dataset size for faster training and helps prevent the model from becoming biased toward the majority class.

1.4.6 Model Evaluation

The system's performance was assessed using standard evaluation metrics: Accuracy to measure the percentage of correctly classified URLs, Precision to determine the proportion of correctly identified phishing URLs, Recall to evaluate the ability to detect all actual phishing URLs, and F1-Score as a balanced measure combining precision and recall.

1.4.7 Model Implementation

Model Implementation was done using python where the user is enabled to input data in the user interface and predict to obtain result in real time.

1.4.7 Tool-Kit

Google Collaboratory (Cloud-Based Jupyter IDE): Provides a free, cloud-based Python environment for developing, training, and testing the machine learning model.

1.5 Significance of Study

This study is significant for several reasons:

Enables real-time detection of phishing attacks: The system's ability to instantly analyse and classify URLs provides immediate protection against active phishing attempts, significantly reducing the window of vulnerability for potential victims.

Provides a lightweight solution suitable for resource-constrained environments: By utilizing efficient algorithms and optimized feature extraction, the system can operate effectively on mobile devices, low-end hardware, and in scenarios with limited computational resources.

Enhances online security with rapid threat mitigation: The quick response time of the detection system allows for timely warnings and preventive measures, minimizing potential damage from phishing attempts before they can cause harm.

Contributes to more efficient and effective phishing detection: The research advances current methodologies by combining machine learning techniques with real-time processing capabilities, offering improved accuracy over traditional detection methods while maintaining computational efficiency.

1.6 Scope of Study and Limitation

The scope of this study encompasses the following aspects:

Data Collection and preprocessing; Gathering large dataset from kaggle of legitimate and phishing URL and prepare it for analysis.

Model development: Uses Logistic Regression and Naive Bayes for fast, efficient detection.

Performance Evaluation: Using metrics such as accuracy, precision, recall, and F1-score to evaluate the models' performance and conducting validation to ensure reliability.

Comparison and Analysis: Comparing the performance of Logistic Regression and Naive Bayes to determine the most effective approach for URL phishing detection.

It covers only URL-based phishing attacks as it does not perform context analysis, it only uses URL- based features as a lexical- centric approach

While the study has the potential to make significant contributions, it also faces several limitations:

Data Availability – The model's performance depends on the quality and diversity of the phishing/legitimate Uniform Resource Locator URL datasets used for training, which may not cover all emerging attack patterns.

Model Generalization – While effective for common phishing tactics, the system may struggle with highly novel or evolving attack methods not seen during training.

Interpretability – Logistic Regression offers some explainability, but certain feature interactions in Uniform Resource Locators URLs may still appear as "black-box" decisions to end-users.

Integration Challenges – Real-world deployment may require adjustments to work seamlessly with existing security tools, browsers, or network architectures

1.7. Definition of terms

In this section, key terms related to the study on machine learning models for detection of URL-based phishing attacks are defined to provide clarity and context.

Phishing: A cybercrime where attackers impersonate legitimate organizations through fake websites or emails to steal sensitive information like passwords or credit card details. Phishing often uses deceptive URLs that mimic real sites.

Uniform Resource Locator (URL): The complete web address used to access online resources, consisting of protocol (https://), domain name, path, and parameters. Phishers manipulate URLs to create malicious links.

Machine Learning: A branch of artificial intelligence where systems learn patterns from data without explicit programming, used to detect phishing by analyzing URL characteristics.

Feature Extraction: The process of converting raw data like URL strings into measurable characteristics for analysis, such as URL length, special characters, or keyword presence.

Tokenization: The process of breaking down text or URLs into smaller units called tokens, which helps in analyzing individual components of web addresses.

Stemming: A text normalization technique that reduces words to their base form by removing prefixes/suffixes, helping group similar words like "phishing" and "phished".

Lemmatization: A more sophisticated alternative to stemming that converts words to their dictionary base form using vocabulary and morphological analysis.

Bag-of-Words: A text representation method that counts word occurrences while ignoring grammar and word order, commonly used in URL analysis.

Chi-squared Test: A statistical method used to determine the relationship between categorical variables, helping identify which URL features best predict phishing.

Normalization: The process of scaling numerical features to a standard range to ensure equal contribution during machine learning model training.

Undersampling: A technique to address class imbalance by reducing the number of samples from the majority class (legitimate URLs) to match the minority class.

Oversampling: The opposite of undersampling, increasing samples from the minority class (phishing URLs) to balance the dataset, often using synthetic data

Synthetic Minority Oversampling Technique (SMOTE): A specific oversampling technique that creates synthetic examples of the minority class rather than simply duplicating existing samples.

Cross-validation: A model evaluation method that partitions data into subsets, using some for training and others for validation to assess performance.

Hyperparameters: Configuration settings for machine learning algorithms that are set before training and affect how models learn from data.

Grid Search: A systematic method for hyperparameter tuning that exhaustively tests all possible combinations of given parameter values.

Logistic Regression: A statistical model used for binary classification problems that predicts the probability of an outcome using a logistic function.

Naive Bayes: A probabilistic classifier based on Bayes' theorem with strong independence assumptions between features, known for its simplicity and speed.

Precision: A metric measuring the accuracy of positive predictions, calculated as true positives divided by all predicted positives.

Recall: A metric measuring the ability to find all positive instances, calculated as true positives divided by actual positives.

F1-score: A harmonic mean of precision and recall that provides a balanced measure of a model's accuracy, especially useful for imbalanced datasets.

ROC Curve: A graphical plot illustrating the diagnostic ability of a binary classifier by showing its true positive rate against false positive rate.

Blacklist: A database containing known malicious URLs or IP addresses used to block access to harmful websites.

HTTPS: A secure communication protocol that encrypts data between web browsers and servers, with its absence potentially indicating phishing attempts.

Whitelist: A database of approved and trusted URLs or IP addresses that are allowed to bypass security filters

AUC: The area under the ROC curve, providing an aggregate measure of model performance across all classification thresholds.

1.8 Organization of the Project

This research project is organized into five chapters:

Chapter One: Presents the introduction to the research, including the background of the study, statement of the problem, aim and objectives, methodology, significance of the study, scope and limitations, and definition of terms.

Chapter Two: Provides a detailed literature review on URL-based phishing attacks, machine learning models for detection, and existing methodologies, along with theoretical frameworks and comparative analyses of related works.

Chapter Three: Describes the research methodology, covering system architecture, data collection and preprocessing, feature engineering, model selection, training, and evaluation metrics.

Chapter Four: Focuses on system implementation, presenting the results of model performance, hyperparameter tuning, challenges encountered, and mitigation strategies.

Chapter Five: Concludes the study with a summary of findings, recommendations for future research, and the overall conclusion of the project.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter provides a comprehensive review of existing literature on machine learning models for phishing detection, with particular emphasis on Uniform Resource Locator (URL)-based approaches. Before delving into the technical methodologies, it is essential to understand the nature and scope of phishing as a cybersecurity threat that has evolved significantly over the past two decades.

2.1.1 Understanding Phishing: Definition and Evolution

Phishing represents one of the most persistent and evolving cybersecurity threats in the digital age. Defined as a form of social engineering attack where attackers impersonate legitimate entities to deceive victims into revealing sensitive information such as passwords, credit card numbers, or personal data, phishing has grown from simple email-based scams to sophisticated, multi-vector attacks that exploit various digital communication channels (Anti-Phishing Working Group, 2023).

The evolution of phishing attacks has been marked by increasing sophistication and diversification of attack vectors. Early phishing campaigns in the late 1990s primarily relied on mass email distribution with obvious grammatical errors and crude website replicas (Moore & Clayton, 2021). However, modern phishing attacks demonstrate remarkable sophistication, employing advanced social engineering techniques, perfect visual replicas of legitimate websites, and targeted campaigns that exploit specific organizational or individual vulnerabilities (Chiew et al., 2020).

Statistical analysis reveals the alarming scale of the phishing threat. The Anti-Phishing Working Group (2023) reported that phishing attacks increased by 61% in 2023, with over 1.27 million unique phishing attacks recorded globally. Financial services remain the most targeted sector, accounting for 34% of all phishing attacks, followed by webmail and Software-as-a-Service (SaaS) platforms at 19% and 15% respectively (Symantec, 2024). The economic impact is equally concerning, with the FBI's Internet Crime Complaint Center reporting over $54 billion in losses attributed to phishing and related social engineering attacks in 2023 (FBI, 2024).

2.2 Theoretical Framework for Phishing Detection

The theoretical foundation of phishing detection systems is built upon pattern recognition principles, statistical learning theory, and cybersecurity frameworks. Traditional approaches to phishing detection were grounded in signature-based recognition, where known malicious patterns were stored in databases for comparison with incoming URLs. However, this approach proved inadequate against the rapidly evolving nature of phishing attacks (Moore & Clayton, 2021).

The transition to machine learning-based detection systems represents a paradigm shift toward predictive modelling, where systems learn to identify potential threats based on feature patterns rather than explicit signatures. This approach is theoretically grounded in statistical learning theory - a mathematical framework that provides the foundation for understanding how algorithms can generalize from training data to make accurate predictions on unseen instances (Vapnik, 2023). Statistical learning theory helps us understand the trade-off between model complexity and generalization ability, providing theoretical guarantees about algorithm performance.

The conceptual framework for URL-based phishing detection encompasses several key theoretical constructs:

Pattern Recognition Theory: Phishing URLs exhibit characteristic patterns that distinguish them from legitimate URLs. These patterns can be captured through feature extraction techniques that transform raw URL strings into numerical representations suitable for machine learning algorithms (Bishop, 2020).

Feature Engineering Theory: The effectiveness of machine learning models depends heavily on the quality and relevance of extracted features. Theoretical work in feature engineering has established principles for selecting and combining features that maximize discriminative power while minimizing computational overhead (Guyon & Elisseeff, 2023).

Classification Theory: URL phishing detection is fundamentally a binary classification problem, where URLs must be categorized as either legitimate or malicious. Various classification algorithms offer different theoretical advantages, from the probabilistic foundations of Naive Bayes to the optimization principles underlying logistic regression (Hastie et al., 2020).

2.3 Evolution of Phishing Detection Methods

The development of phishing detection systems has progressed through distinct evolutionary phases, each characterized by different technological approaches and theoretical foundations.

2.3.1 First Generation: Signature-Based Systems (2000-2010)

Early phishing detection systems relied heavily on blacklist databases and signature matching techniques. These systems maintained repositories of known malicious URLs and used exact or near-exact matching to identify threats. Signature-based detection works by maintaining a database of known malicious patterns or exact URLs and comparing incoming requests against this database for matches. Symantec (2020) reported that first-generation systems achieved detection rates of 60-70%, but suffered from significant limitations in detecting zero-day attacks.

The reactive nature of these systems meant they were always one step behind attackers, who could easily evade detection by modifying URL structures or using previously unknown domains (Anti-Phishing Working Group, 2022).

2.3.2 Second Generation: Heuristic Rule-Based Systems (2005-2015)

The limitations of pure signature-based approaches led to the development of heuristic rule-based systems that incorporated domain expertise into automated detection rules. Heuristic-based detection involves creating rules based on expert knowledge about phishing characteristics, such as "URLs longer than 75 characters are suspicious" or "domains registered less than 30 days ago are high-risk." These systems analyzed various URL characteristics, including length, special character usage, domain age, and SSL certificate validity (Chiew et al., 2020).

However, these systems suffered from high maintenance overhead and difficulty in adapting to new attack techniques (Wang et al., 2023).

2.3.3 Third Generation: Machine Learning-Based Systems (2010-Present)

The emergence of machine learning approaches marked a fundamental shift in phishing detection methodology. Machine learning-based detection uses algorithms that automatically learn patterns from training data to distinguish between legitimate and phishing URLs, without requiring explicit programming of detection rules.

This transition was enabled by several developments:

Availability of Large Datasets: The growth of the internet made it possible to assemble large, labeled datasets of URLs suitable for machine learning training (Vrbančič et al., 2020).

Advances in Feature Extraction: Researchers developed sophisticated techniques for extracting meaningful features from URLs (Sahingoz et al., 2020).

Algorithmic Improvements: More powerful and efficient machine learning algorithms provided the computational tools necessary for real-time phishing detection (Hastie et al., 2020).

2.4 Machine Learning Algorithms in Phishing Detection

Machine learning is a branch of artificial intelligence that enables systems to automatically learn patterns from data and make predictions without being explicitly programmed. In the context of phishing detection, it empowers models to identify suspicious URLs by analyzing their structural, lexical, and semantic features, and by generalizing from patterns observed in past phishing and legitimate examples. Algorithms in phishing detection such as Logistic Regression, Multinomial Naive Bayes, and Random Forest have demonstrated strong performance in accurately classifying URLs. These algorithms differ in their approach ranging from statistical probability estimation to ensemble-based decision-making but share the common goal of improving detection accuracy, interpretability, and computational efficiency for real-time security applications (Kumar et al., 2023).

2.4.1 Logistic Regression in Phishing Detection

Logistic Regression is a statistical method used for binary classification that models the probability of an outcome using the logistic function. Unlike linear regression which predicts continuous values, logistic regression predicts the probability that an instance belongs to a particular category (in this case, phishing or legitimate).

The mathematical foundation of logistic regression in phishing detection can be expressed as it is in the Equation 2.1:

$$Y=\beta_0+\beta_1 X_1+\beta_2 X_2+\ldots+\beta_n X_n \quad (2.1)$$

Where Y represents the binary outcome (phishing or legitimate), X represents the feature vector extracted from URLs, and β coefficients represent the learned weights.

Kumar et al. (2023) conducted a comprehensive evaluation of logistic regression against other classification algorithms using a balanced dataset of URLs. Their study found that logistic regression achieved 97% accuracy while maintaining excellent interpretability, making it particularly valuable for security applications where understanding decision factors is crucial.

Theoretical advantages include:

Probabilistic Output: Provides probability scores for threat ranking

Computational Efficiency: Fast inference times crucial for real-time applications

Interpretability: Coefficient weights provide insights into feature importance

Mahajan and Siddavatam (2023) investigated regularization techniques - methods to prevent overfitting by adding penalty terms to the loss function. Regularization helps models generalize better by preventing them from becoming too complex. Their study demonstrated that L2 regularization (Ridge regression) consistently outperformed L1 regularization (Lasso) for phishing detection tasks, achieving 97.14% accuracy on a dataset of 36,711 URLs.

2.4.2 Multinomial Naive Bayes in Phishing Detection

Multinomial Naive Bayes is a probabilistic classifier based on Bayes' theorem with a "naive" assumption that features are conditionally independent. Despite this unrealistic assumption, it often performs well in practice, especially for text classification tasks. The algorithm is particularly suited for discrete features like word counts or token frequencies in URLs.

The mathematical formulation can be expressed as: $P(C \vee X) \propto P(C) \prod_{I=1}^{|V|} P(x_i \vee C)^{X_i}$

(2.2)

Where C represents the class (legitimate or phishing), X represents the feature vector of token counts from URL analysis, and x_i represents the count of token i in the URL, as seen in equation 2.2

Alam et al. (2023) conducted an extensive evaluation of Multinomial Naive Bayes for phishing detection, demonstrating its particular strength in analyzing textual components of URLs. Their research showed that MNB achieved 97% accuracy when applied to tokenized URL features.

Theoretical advantages include:

Text-Oriented Design: Naturally suited for analyzing URL strings as collections of tokens

Handling High-Dimensional Sparse Features: Efficiently handles URL feature vectors with many zero values

Computational Efficiency: Linear time complexity in both training and prediction

Robustness to Irrelevant Features: The independence assumption prevents irrelevant features from significantly impacting decisions

2.4.3 Random Forest in Phishing Detection

Random Forest is an ensemble learning method that combines multiple decision trees trained on different subsets of data and features. Each tree votes for a classification, and the final prediction is made by majority vote. This approach reduces overfitting and improves generalization compared to individual decision trees.

Mohammad et al. (2024) conducted one of the most comprehensive evaluations of Random Forest for phishing detection, using a dataset of over 100,000 URLs with 30 distinct features. Their research demonstrated Random Forest achieving 98.3% accuracy, outperforming individual decision trees by 5.2% and logistic regression by 2.1%.

Ensemble learning refers to methods that combine multiple learning algorithms to obtain better predictive performance than could be obtained from any of the constituent learning algorithms alone. Random Forest exemplifies this by combining many decision trees.

The ensemble approach addresses several challenges:

Overfitting Prevention: Multiple trees trained on different data subsets reduce overfitting risk

Feature Importance Analysis: Naturally provides measures of which features are most important

Handling Mixed Feature Types: Effectively handles numerical, categorical, and binary features without preprocessing

2.5 Advanced Feature Engineering Techniques

Feature engineering represents a critical component in machine learning-based phishing detection. Feature engineering is the process of selecting, modifying, or creating features from raw data to improve machine learning model performance.

2.5.1 Lexical Feature Analysis

Lexical features focus on the textual and structural characteristics of URLs that can be analyzed without requiring external resources. These features form the foundation of lightweight detection systems suitable for real-time deployment.

Jitendra Kumar et al. (2023) conducted extensive analysis of lexical features, examining over 20 distinct URL characteristics. Their research identified several key indicators:

URL Length Distribution: URLs exceeding 75 characters showed 73% correlation with phishing classification

Special Character Analysis: Phishing URLs showed 2.3 times higher usage of hyphens compared to legitimate sites

Digit Integration Patterns: Phishing sites showed 45% higher digit usage in domain names

Hannousse and Yahiouche (2021) extended lexical analysis to include entropy-based features - measures of randomness in character sequences. Entropy quantifies the unpredictability of information content; higher entropy indicates more randomness. Algorithmically generated phishing URLs exhibited significantly higher entropy values (average 4.2) compared to legitimate URLs (average 2.8).

2.5.2 Domain-Based Feature Engineering

Domain-based features analyze characteristics of the domain component of URLs, including top-level domains, subdomain structures, and domain age information.

Geng et al. (2020) developed a comprehensive framework examining:

Top-Level Domain Analysis: Certain TLDs (.tk, .ml, .ga) showed 8-12 times higher phishing rates

Subdomain Complexity: URLs with more than three subdomain levels showed 67% correlation with phishing

Domain Age Correlation: Newly registered domains (less than 30 days) accounted for 78% of phishing sites

2.5.3 Semantic Feature Extraction

Semantic features attempt to capture the meaning and intent behind URL construction through analysis of keywords, brand names, and suspicious phrases.

Tyagi et al. (2020) developed semantic analysis techniques including:

Keyword Density Analysis: Identification of suspicious keywords (login, verify, update, secure) that appear disproportionately in phishing URLs

Brand Name Detection: Automated identification of legitimate brand names appearing in suspicious contexts

Language Model Analysis: Application of statistical language models to identify URLs deviating from typical naming conventions

2.6 Deep Learning Integration Frameworks

Deep Learning Integration Frameworks refer to approaches that combine advanced deep learning models with traditional machine learning techniques to enhance phishing detection.

Figure 2.1 shows the diagrammatic representation of deep learning frame work.

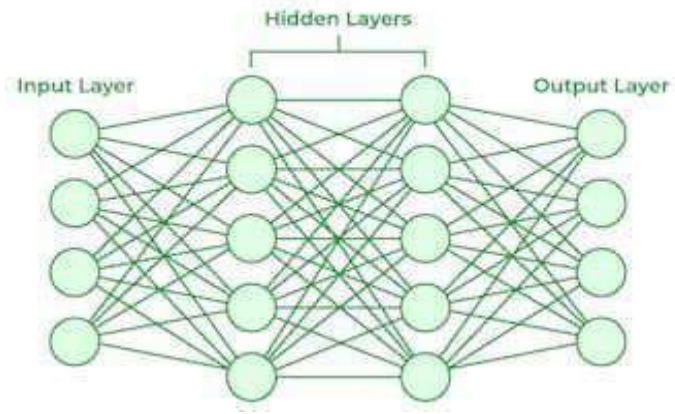


Figure 2.1: Deep Learning Integration Framework

Deep learning is a subset of machine learning that uses multi-layered artificial neural networks to automatically learn complex patterns and representations from data, often achieving superior performance without extensive manual feature engineering, Li et al. (2023)

2.6.1 Hybrid CNN-Traditional ML Approaches

Convolutional Neural Networks (CNNs) are deep learning models originally designed for image processing but adaptable to sequence data like URLs. CNNs use convolutional layers to automatically detect local patterns and features in data.

Li et al. (2023) pioneered a novel hybrid architecture combining CNNs for automatic feature extraction with Random Forest for final classification. This hybrid approach combines the automatic feature discovery capabilities of deep learning with the interpretability and efficiency of traditional machine learning.

The architecture operated through:

Feature Extraction Stage: CNN analyzed URL strings using varying filter sizes to capture different pattern granularities

Classification Stage: Random Forest provided final classification with interpretable feature importance

This approach achieved 98.5% accuracy while maintaining interpretability through feature importance analysis.

2.6.2 Transfer Learning Applications

Transfer Learning is a machine learning technique where a model developed for one task is reused as the starting point for a related task. This leverages knowledge gained from pre-training on large datasets to improve performance on specific tasks with limited data.

Wang et al. (2023) explored transfer learning using pre-trained Bidirectional Encoder Representations from Transformers (BERT) models for phishing detection. BERT is a transformer-based model pre-trained on large text corpora that understands contextual relationships in language. The framework provided advantages:

Reduced Training Data Requirements: Achieved competitive performance with 60% less training data

Multilingual Capabilities: Pre-trained multilingual models enabled detection across various languages

Contextual Understanding: BERT's attention mechanisms understood contextual relationships between URL parts

The approach achieved 98.8% accuracy while demonstrating strength in detecting culturally-specific phishing patterns.

2.7 Specialized Detection Systems

Mobile-optimized systems increasingly leverage on-device machine learning and lightweight.

2.7.1 Mobile-Optimized Detection Systems

The proliferation of mobile internet usage has driven research into mobile-optimized phishing detection systems that balance effectiveness with computational constraints.

Okafor et al. (2023) developed a lightweight Random Forest variant optimized for Android devices:

Memory Optimization: Reduced memory footprint to less than 10MB while maintaining 94.2% accuracy

Battery Efficiency: Reduced battery consumption by 40% compared to standard implementations

Offline Capability: Operated entirely offline, eliminating data usage costs

Real-Time Performance: Achieved sub-100ms detection latency

2.7.2 Cultural and Regional Adaptations

Adebayo et al. (2023) developed a culturally-aware phishing detection system for Nigerian internet users:

Local Language Integration: Incorporated Nigerian Pidgin English and major local languages

Cultural Brand Awareness: Enhanced recognition of local institutions commonly spoofed

Infrastructure Constraints: Prioritized lightweight algorithms for resource-limited environments

Mobile-First Design: Optimized for smartphone deployment with data usage minimization

2.8 Performance Optimization and Deployment Considerations

Deployment strategies often favour hybrid models that use fast lexical features for initial filtering and more complex, slower models for in-depth analysis of high-risk candidates.

2.8.1 Real-Time Processing Requirements

Real-time processing refers to systems that can process data and provide responses within strict time constraints, typically measured in milliseconds for web applications.

Liu et al. (2024) analyzed processing latency across different algorithms:

Feature Extraction Overhead: Lexical features required 5-15ms per URL, while content-based analysis exceeded 200ms

Algorithm Inference Time: Logistic Regression and Naive Bayes showed sub-millisecond inference times

Memory Access Patterns: Linear models showed superior performance due to sequential memory access

Parallelization Potential: Ensemble methods offered excellent scaling across multiple CPU cores

2.8.2 AutoML Framework

AutoML (Automated Machine Learning) refers to systems that automate the machine learning pipeline, including algorithm selection, hyperparameter tuning, and model deployment.

Kumar et al. (2023) developed an AutoML framework incorporating:

Automated Model Selection: Continuously evaluated multiple algorithms for optimal performance

Hyperparameter Optimization: Used Bayesian optimization - a technique for efficiently finding optimal hyperparameters by building a probabilistic model of the objective function

Incremental Learning: Supported online learning capabilities for adapting to new attack patterns

Resource Monitoring: Comprehensive monitoring of computational resources and performance metrics

2.9 Emerging Research Directions

Emerging research is now focusing on generative adversarial networks (GANs) to proactively create and defend against novel, evolving adversarial attack strategies.

2.9.1 Adversarial Robustness

Adversarial attacks are inputs specifically designed to fool machine learning models by making small, often imperceptible changes that cause misclassification.

Martinez et al. (2024) developed adversarial training techniques addressing:

Evasion Attacks: Attackers modify phishing URLs to fool detection systems while maintaining malicious functionality

Poisoning Attacks: Attempts to compromise training data with carefully crafted examples to degrade model performance

Model Extraction: Attacks aimed at reverse-engineering detection models to understand decision boundaries

Adversarial training involves training models on both normal and adversarially, which modified examples to improve robustness. Their techniques improved robustness against evasion attacks by 34%.

2.9.2 Privacy-Preserving Detection

Federated Learning is a distributed machine learning approach where multiple parties collaboratively train a shared model without sharing their raw data. Each participant trains a local model on their private data, then shares only model parameters (not data) with a central coordinator that aggregates updates to improve the global model.

Zhang et al. (2023) pioneered federated learning for phishing detection, addressing:

Data Localization: Training data never left local devices or institutional boundaries

Differential Privacy: Mathematical guarantees of individual privacy protection during collaborative training

Secure Aggregation: Cryptographic techniques protected individual model updates during aggregation

Performance Maintenance: Despite privacy constraints, achieved 96.2% accuracy

Differential Privacy is a mathematical framework that provides quantifiable privacy guarantees by adding carefully calibrated noise to data or query results, ensuring that individual records cannot be distinguished.

2.10 Interpretability and Explainability

As machine learning systems become more prevalent in security applications, the need for interpretable and explainable decisions has grown increasingly important.

Explainable AI (XAI) refers to methods and techniques that make the results of AI systems understandable to humans. This is crucial in cybersecurity where analysts need to understand why certain decisions were made.

Thompson et al. (2025) explored integration of explainable AI techniques with phishing detection systems, addressing:

Regulatory Compliance: Financial and healthcare regulations increasingly require explanations for automated security decisions

Security Analyst Support: Human analysts need to understand why specific URLs were flagged

System Debugging: Interpretable models facilitate identification of systematic errors or biases

User Trust: End-users are more likely to trust systems that can explain their decisions

LIME (Local Interpretable Model-agnostic Explanations) is a technique that explains individual predictions by approximating the model locally with an interpretable model. For phishing detection, LIME can highlight which specific URL features contributed most to classification decisions.

The research developed LIME integration techniques providing feature-level explanations for phishing detection decisions while maintaining 95.9% accuracy.

2.11 Comparative Algorithm Performance Studies

Algorithm performance is often selected for high-stakes deployments where accuracy is paramount, despite its greater computational cost.

2.11.1 Large-Scale Comparative Analyses

Korkmaz et al. (2023) conducted extensive comparative evaluation analyzing eight different machine learning algorithms across three datasets totalling over 200,000 URLs. Their methodology employed rigorous statistical validation including stratified cross-validation a validation technique that ensures each fold of the data maintains the same proportion of samples for each target class. findings achieved the following:

Random Forest Dominance: Achieved highest average accuracy (97.2%) across all datasets

Naive Bayes Efficiency: Achieved 95.8% accuracy with inference times 3-5 times faster than ensemble methods

SVM Inconsistency: Support Vector Machines showed high variance (89.3% to 96.7% accuracy) across datasets

Deep Learning Overhead: Neural networks achieved competitive accuracy (96.1%) but required significantly more computational resources

2.11.2 Feature-Specific Performance Analysis

Subasi et al. (2023) analyzed algorithm performance across different feature types:

Lexical Feature Specialists: Logistic Regression and Multinomial Naive Bayes achieved 94.5% and 93.8% accuracy using only URL string analysis

Ensemble Advantage: Random Forest demonstrated superior performance (97.8%) when combining multiple feature types

Content Feature Requirements: Algorithms dependent on content analysis showed degraded performance when limited to URL-only features.

2.12 Challenges and Limitations

These limitations necessitate the use of sophisticated data augmentation and continuous learning pipelines to maintain model efficacy in production environments.

2.12.1 Dataset Quality and Bias Issues

Vrbančič et al. (2020) identified critical issues with commonly used phishing detection datasets:

Temporal Bias: Datasets over-represent techniques from specific time periods

Geographic Bias: Under-representation of attacks targeting specific regions

Labeling Inconsistencies: Inter-annotator agreement rates sometimes below 85%

Class Imbalance Evolution: Artificial balance may not reflect real-world deployment conditions.

2.12.2 Computational Constraints

Real-world deployment introduces constraints often overlooked in research:

Memory Limitations: Mobile and edge devices have strict memory constraints

Processing Latency: Real-time applications require sub-second response times

Power Consumption: Battery-powered devices need energy-efficient algorithms

Network Bandwidth: Limited connectivity affects feature extraction capabilities

2.13 Research Gaps and Opportunities

Based on comprehensive literature review, several significant research gaps have been identified:

Lightweight Real Time Detection Systems: Most existing research focuses primarily on accuracy metrics while giving insufficient attention to inference time, memory usage, and power consumption. This gap is particularly significant for mobile and edge computing scenarios.

Comprehensive Feature Engineering for URL-Only Detection: Many studies incorporate features requiring external data sources (WHOIS information, content analysis) that may not be available in real-time deployment scenarios. There is need for sophisticated lexical and structural feature extraction operating entirely on URL strings.

Comparative Analysis of Lightweight Algorithms: Most comparative studies focus on accuracy without adequate consideration of computational efficiency and deployment feasibility. Comprehensive evaluation considering both effectiveness and efficiency metrics is needed.

2.14 Conclusion

The analysis reveals that while significant progress has been made in achieving high detection accuracy, considerable opportunities remain for developing lightweight, efficient systems suitable for real-time deployment in resource-constrained environments. The focus on URL-only detection using computationally efficient algorithms represents a promising approach to balancing security effectiveness with practical deployment requirements.

The comparative analysis demonstrates the continued relevance of traditional machine learning approaches like Logistic Regression and Multinomial Naive Bayes, particularly when optimized for specific deployment scenarios. These algorithms offer advantages in interpretability, computational efficiency, and robustness that make them well-suited for security applications.

Future research directions identified suggest continued evolution toward more adaptive, privacy-preserving, and culturally-aware detection systems. The integration of traditional machine learning with emerging techniques like federated learning and adversarial training represents promising avenues for developing next-generation phishing detection systems.

2.15 Summary of Related Work

The Table 2.1 shows the summary of related work

Authors	Work Title	Algorithm(s)	Dataset Size	Accuracy (%)	Key Features	Primary Contribution
Kumar et al.(2023)	Feature-Driven Phishing Detection with Interpretable ML	LR, RF, SVM	50,000 URLs	LR: 97%, RF: 95.8%	25 lexical & domain features	Comprehensive feature analysis with interpretability
Alam et al. (2023)	NLP-Based Phishing URL Classification	MNB, RF, SVM	36,711 URLs	MNB: 97%, RF: 97%	Tokenized URL features	Text-based URL analysis with NLP techniques
Mohammad et al. (2024)	Large-Scale Benchmarking of Phishing Detection Models	RF, DT, LR, SVM, NB	100,000+ URLs	RF: 98.3%, LR: 96.2%	30 mixed feature types	Large-scale comparative evaluation
Korkmaz et al.(2023)	Statistical Validation of ML for Phishing Detection	8 ML algorithms	200,000+ URLs	RF: 97.2%, NB: 95.8%	Comprehensive feature set	Statistical validation of performance
Subasi et al.(2023)	Algorithm-Feature Interactions in Phishing Detection	LR, MNB, RF, SVM	75,000 URLs	MNB: 96.5%, LR: 94.5%	Feature-type analysis	Algorithm-feature interaction analysis
Mahajan & Siddavatam (2023)	Regularized Models for High-Dimensional	Regularized LR	36,711 URLs	Ridge: 97.14%	High-dimensional features	Regularization for overfitting prevention

	Phishing Data					
Li et al. (2023)	Hybrid Deep Learning for Automated Phishing Detection	Hybrid CNN-RF	150,000 URLs	98.5%	Auto-extracted features	Deep learning integration with traditional ML
Wang et al. (2023)	Transfer Learning for Context-Aware Phishing Detection	Transfer Learning	120,000 URLs	98.8%	Contextual embeddings	NLP advances applied to phishing detection
Okafor et al.(2023)	Lightweight RF for Mobile Phishing Detection	Lightweight RF	Mobile dataset	94.2%	Mobile-optimized features	Mobile deployment optimization
Adebayo et al.(2023)	Culturally-Adapted Phishing Detection for Nigerian URLs	Culturally-aware ML	Nigerian dataset	95.8%	Cultural & linguistic features	Region-specific adaptation
Liu et al. (2024)	Real-Time Phishing Detection with Latency Optimization	Multiple algorithms	Real-time dataset	Varies	Latency-optimized features	Real-time processing analysis
Martinez et al.(2024)	Adversarially Robust Phishing URL Classifiers	Adversarially robust	Attack datasets	91.2% (robust)	Adversarial training	Robustness against adversarial attacks
Zhang et al. (2023)	Adversarially Robust Phishing URL Classifiers	Federated Learning	Distributed dataset	96.2%	Privacy-preserving features	Privacy-preserving collaborative learning
Thompson et al.(2025)	Explainable AI for Transparent Phishing	Explainable AI	Interpretability dataset	95.9%	Interpretable features	LIME integration for explainability

	Detection					
Hannousse & Yahiouche (2021)	Entropy-Based Detection of Malicious URLs	Entropy-based analysis	73,036 URLs	94.7%	Entropy & lexical features	Entropy analysis for URL randomness
Geng et al. (2020)	Domain-Centric Phishing Detection with ML	Domain-focused ML	45,000 URLs	93.8%	Domain-based features	Domain characteristic analysis
Tyagi et al. (2020)	Semantic Analysis for Brand Spoofing Detection	Semantic analysis	Brand spoofing dataset	94% detection	Semantic & brand features	Brand impersonation detection
Vrbančič et al.(2020)	Systematic Evaluation of Phishing Dataset Quality	Dataset analysis	Multiple datasets	N/A (analysis)	Dataset quality metrics	Systematic dataset quality evaluation

Table 2.1: Summary of Related Work in Phishing Detection

CHAPTER THREE

METHODOLOGY

3.1 Introduction

This chapter presents a rigorous methodology for developing a machine learning models for detection of URL-based phishing attacks. It encompasses a comprehensive dataset sourced from Kaggle, comprising 549,347 URLs labelled as either "Good" (legitimate) or "Bad" (phishing). The dataset's integrity is ensured by the absence of missing values and a balanced class distribution, making it ideal for training robust models. The methodology follows a systematic pipeline, from data preprocessing to model deployment, with a focus on computational efficiency and real-time applicability. Key innovations include feature engineering tailored to URL structures and the use of interpretable models for transparency in decision-making. The approach prioritizes both security effectiveness and practical deployment considerations, ensuring that the resulting system can be implemented in real-world scenarios with minimal computational overhead.

3.2 Conceptual Framework

The Conceptual framework implements a client-side phishing detection framework as shown in the workflow diagram in figure 3.1. The framework consists of three components:

Attacker: Sends phishing URLs to victims through various channels, attempting to redirect them to malicious websites.

Victim with machine learning (ML) Protection: The target user is protected by a machine learning model deployed locally on their device for real-time threat detection.

Detection Process: When a victim receives and attempts to access a URL, the local ML model analyzes it immediately. Legitimate websites allow normal access, while detected phishing attempts are blocked to prevent compromise.

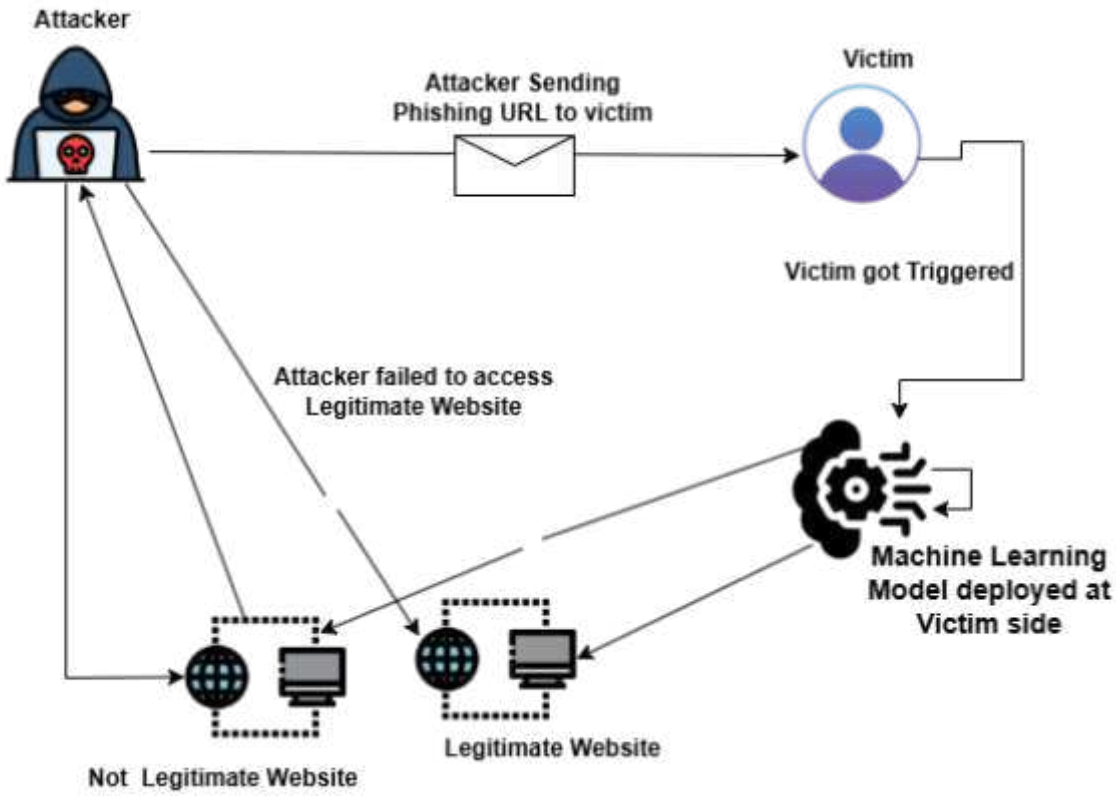


figure 3.1: Conceptual Framework

3.3 System Architecture

The process starts with a raw dataset of URLs (both legitimate and phishing) from Kaggle. The data is preprocessed by cleaning missing values and ensuring consistent formatting. Features were selected and converted into numerical values, then normalized for fairness. The dataset then split into 80% training and 20% testing sets. Our two selected Models Logistic Regression and Multinomial Naive Bayes were trained on the data, followed by hyperparameter tuning using Grid Search and Cross-Validation to find the best settings. The models were evaluated on the test set using metrics: accuracy, precision, Recall and F1-score, the best-performing model was selected as the final classifier for real-time URL analysis and the implementation was done through the interface where the user can input a URL for an instant real time detection.

Figure 3.2 demonstrated the proposed system architecture for URL-based phishing attacks detection.

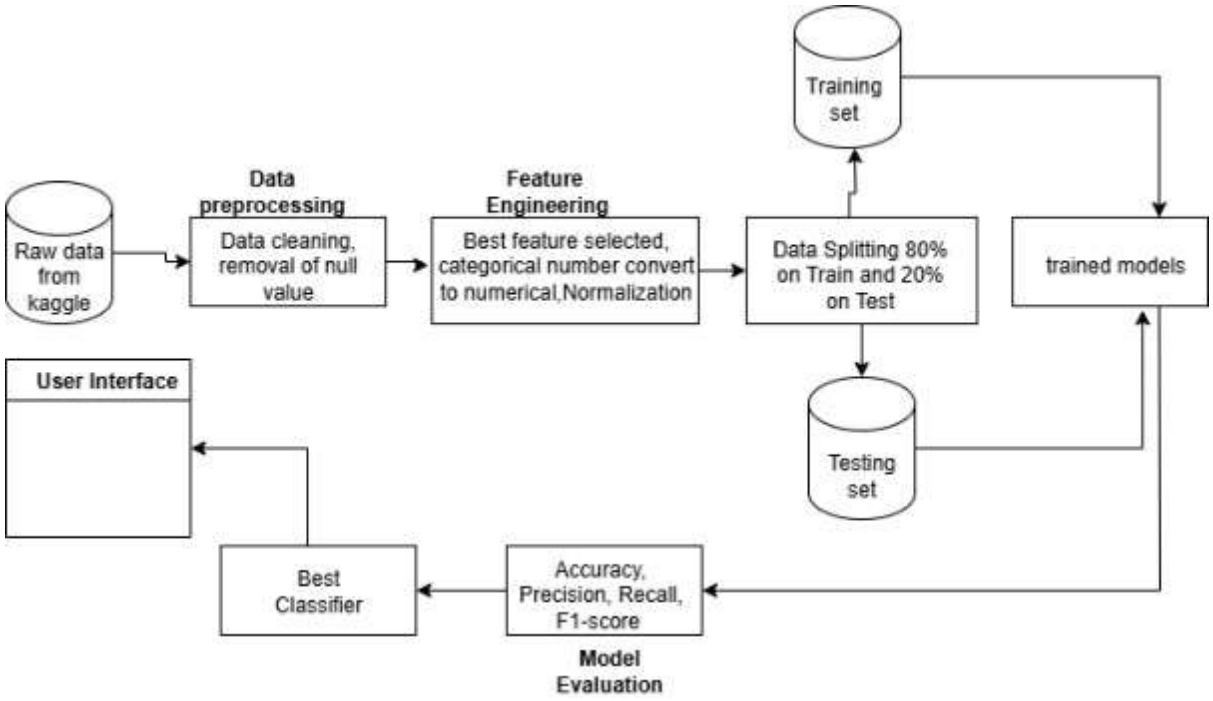


Figure 3.2: System architecture for URL-based phishing attacks detection

3.3 Data Collection

The dataset was obtained from Kaggle curated repository, comprising URLs in complete web address string format and binary classifications labelled as "Good" (legitimate) or "Bad" (phishing).

3.3.1 Data Source

The dataset (phishing_site_urls.csv) was sourced from Kaggle, with link via: (https://www.kaggle.com/datasets/taruntiwarihp/phishing-site-urls) it was chosen for its rigorous review process and community validation. Each URL was accurately labeled, with phishing entries verified through browser security tools and domain reputation checks. Spanning multiple years, the dataset provides patterns across evolving phishing techniques, ensuring a credible foundation for the detection system.

3.3.2 Data Description

The dataset consists of 549,347 entries organized into two columns: the first contains the complete URL string, and the second is the Label column indicating whether the URL is Good (legitimate) or Bad (phishing). Good URLs do not contain malicious content, while Bad URLs are confirmed phishing sites. The dataset has no missing values, ensuring completeness and reliability. It includes a diverse range of domains, protocols (HTTP and HTTPS), path complexities, and query parameters, exposing models to various URL structures.

3.6 Libraries

The implementation relies on several key Python libraries to facilitate data processing, analysis, and model development.

Pandas is used for data manipulation and analysis, especially for loading and processing URL datasets.

NumPy provides essential functionality for numerical computations and array operations that underpin machine learning algorithms.

For data visualization, **Seaborn and Matplotlib** enable statistical visualization and basic plotting respectively, while **Plotly** offers interactive visualizations for dynamic analysis. The Time library helps measure execution time and performance for real-time detection evaluation. The core machine learning implementations utilize Logistic Regression and Multinomial NB from scikit-learn, along with supporting functions like train test split for dataset division and classification report for performance metrics generation.

Text processing leverages **Regexp Tokenizer** for URL tokenization and **SnowballStemmer** for feature normalization.

CountVectorizer transforms text features into numerical vectors suitable for machine learning. Supporting libraries include make pipeline for streamlined processing workflows, Image for visualization display,

WordCloud for frequency visualization, Selenium for web browser automation when needed, Warnings for message management.

These libraries together form a comprehensive toolkit that enables sophisticated phishing detection while maintaining computational efficiency.

3.7 Data Preprocessing

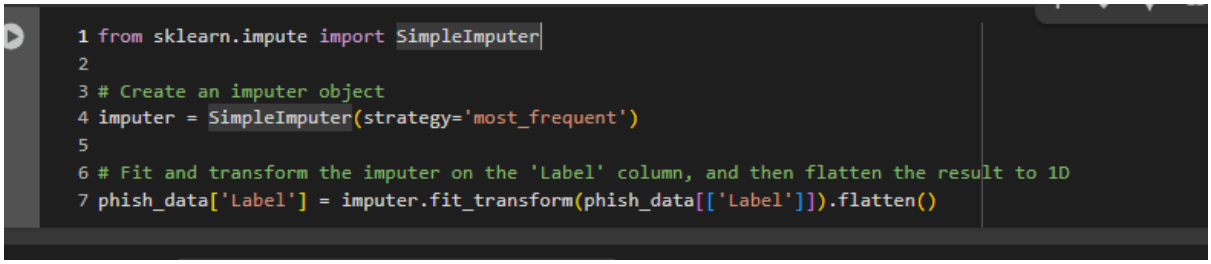
The data preprocessing phase transforms raw URLs into a standardized format suitable for feature extraction and analysis.

3.7.1 Data Cleaning

Data cleaning involved identifying and addressing missing values, duplicates, and inconsistencies within the dataset. Missing values were handled using imputation; fit_ransform() which is performed in two steps: Fit: The imputer learns the most frequent value in the 'Label' column. Transform: It replaces all missing values in the 'Label' column with that most frequent value.

preprocessing is essential to ensure quality and consistency. Preprocessing is crucial to eliminate noise and improve model accuracy.

The figure 3.3 shows how imputer was implemented to handle missing value.



```
from sklearn.impute import SimpleImputer

# Create an imputer object
imputer = SimpleImputer(strategy='most_frequent')

# Fit and transform the imputer on the 'Label' column, and then flatten the result to 1D
phish_data['Label'] = imputer.fit_transform(phish_data[['Label']]).flatten()
```

figure 3.3 Creating Simple Imputer for Missing Value

3.8 Feature Engineering

In the feature engineering two approaches were implemented which were feature extraction and feature selection, It was vital to enable machine learning models to effectively classify phishing URLs. The approaches used combined lexical, host-based, and behavioral features in it feature extraction, with a focus on text-based feature extraction using tokenization, stemming, and vectorization techniques while **Chi-squared (χ^2) test** was essential in handling top correlated variables in the feature selection.

3.8.1 Feature Extraction

To convert URL text into machine-readable features, the following NLP processing tasks were performed, namely; Tokenization with RegexpTokenizer, Normalization with SnowballStemmer, and Vectorization with CountVectorizer.

i. RegexpTokenizer: Uses **regular expressions** to split URLs into tokens (e.g., keeping ".exe", "virus" as single tokens). Helps in isolating high-risk phrases instead of just splitting by spaces. For
instance splitting "www.paypal.com" into [“www”, "paypal", "com"], without losing context.

The figure 3.4 shows how RegexpTokenizer was implementation to help splits URL into tokens

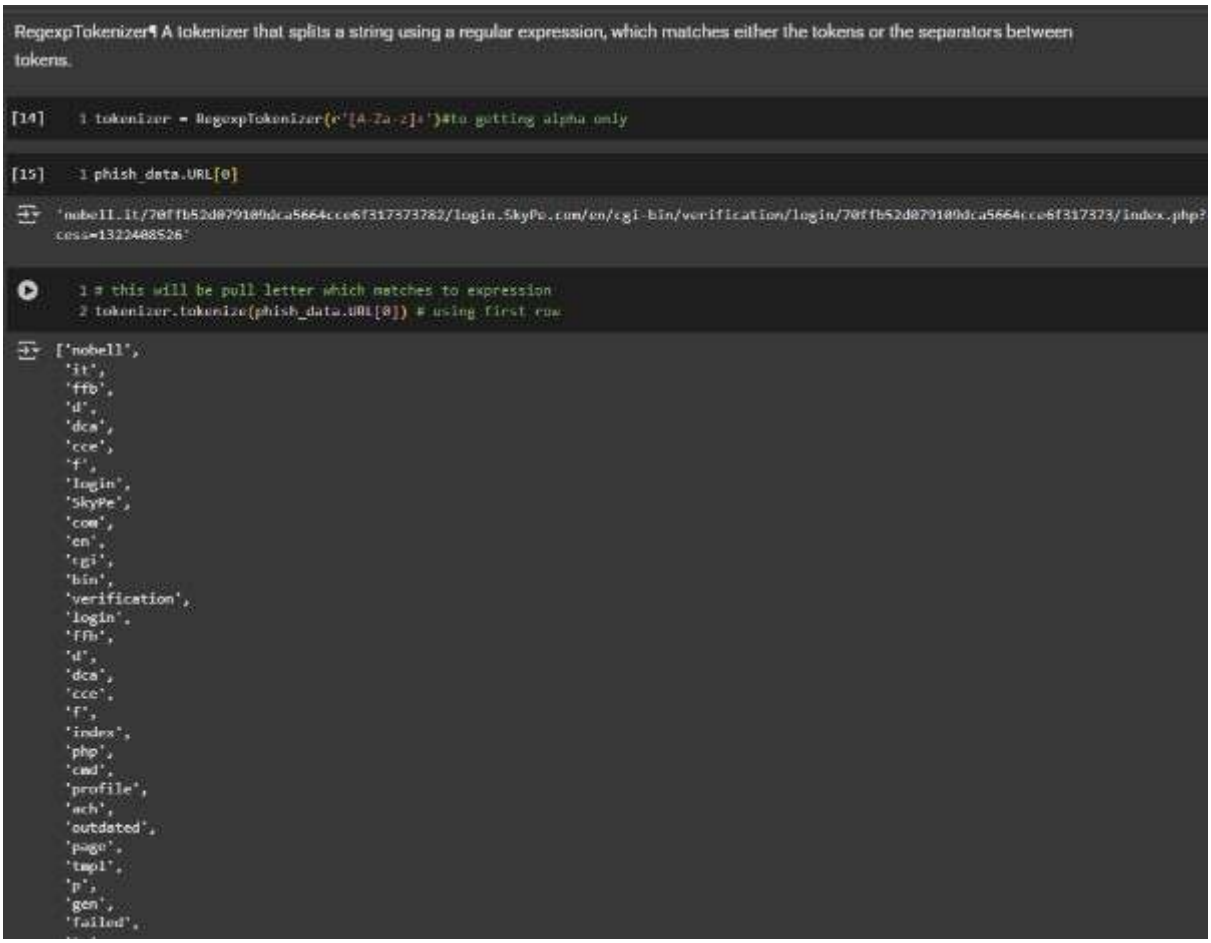


figure 3.4 RegexpTokenizer implementation

ii. SnowballStemmer: Normalize tokens to root forms to reduce redundancy. Converted tokens like "phishing" → "phish", "logged" → "log". This process also Joined stemmed tokens into sentences for vectorization.

The figure 3.5 shows how SnowballStemmer implementation which take tokens to their root form.

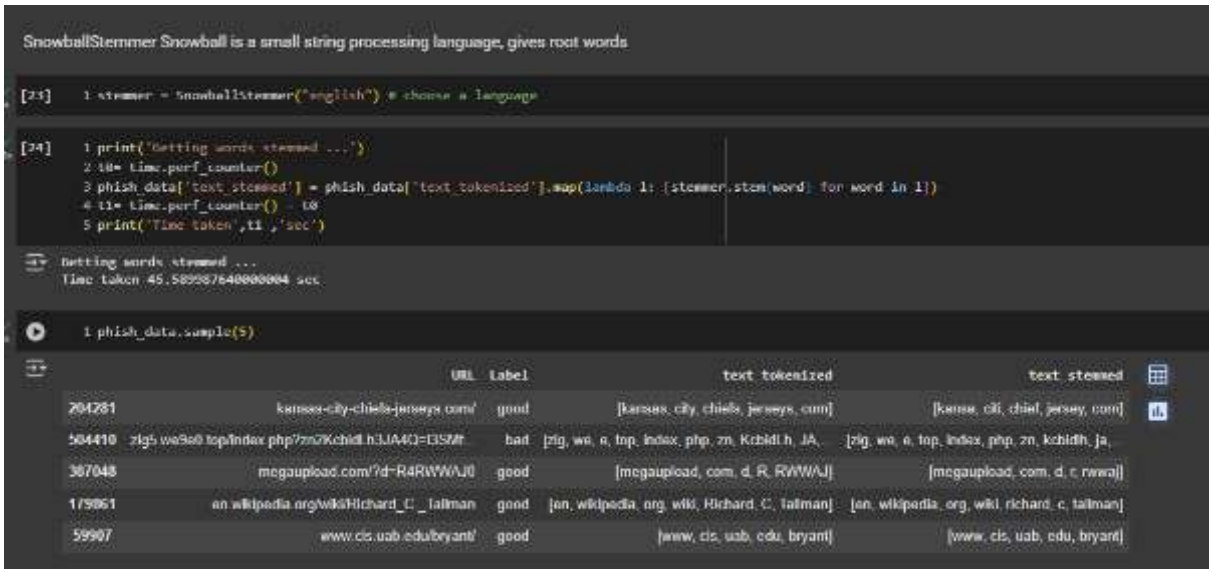


figure 3.5 SnowballStemmer implementation

iii. CountVectorizer (Bag-of-Words): This process help Splits URLs into tokens (words) and counts their occurrences.

This was Performed by Creating a sparse matrix where each column represents a word for example "login" and each row is a URL.

CountVectorizer was quite useful because URLs feature containing meaningful keywords like ("paypal", "login") in the URLs that indicate phishing were Splits into tokens and it was also useful for a numerical representation as categorical representation is usually a problem to train in machine learning (ML) algorithms.

3.8.2 Feature selection was performed using the **Chi-squared (χ^2) test** to determine which words (or tokens) in URLs are most statistically significant for predicting phishing in the Label feature in the dataset. Key features were selected from the URLs to facilitate machine learning analysis and training.

The figure 3.6 shows implementation of Chi-squared test.

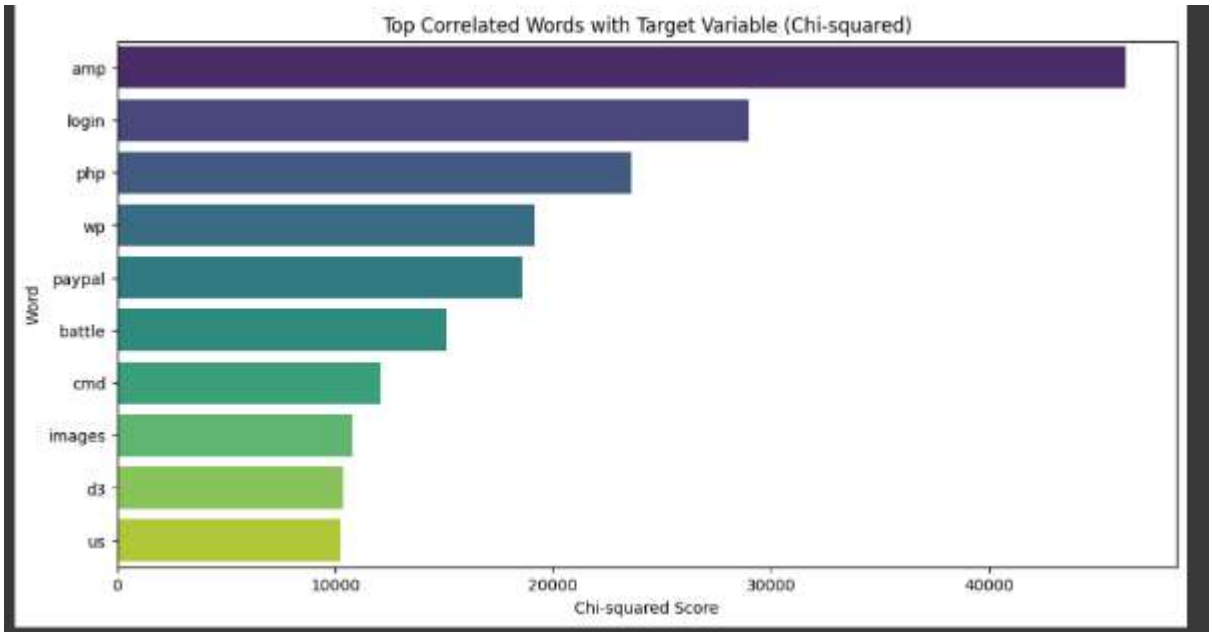


Figure 3.6 Feature selection using Chi-squared (χ^2)

3.9 Data Transformation

Tokenized and stemmed URL components were converted into a numerical feature matrix using a CountVectorizer (Bag-of-Words) approach. Tokens produced by RegexpTokenizer were stemmed with SnowballStemmer, joined back into strings, and then vectorized with CountVectorizer to produce a high-dimensional sparse matrix (each column = token count). This matrix was used as input to the Logistic Regression and Multinomial Naive Bayes models.

In the figure 3.7, It shows processes of data transformation.

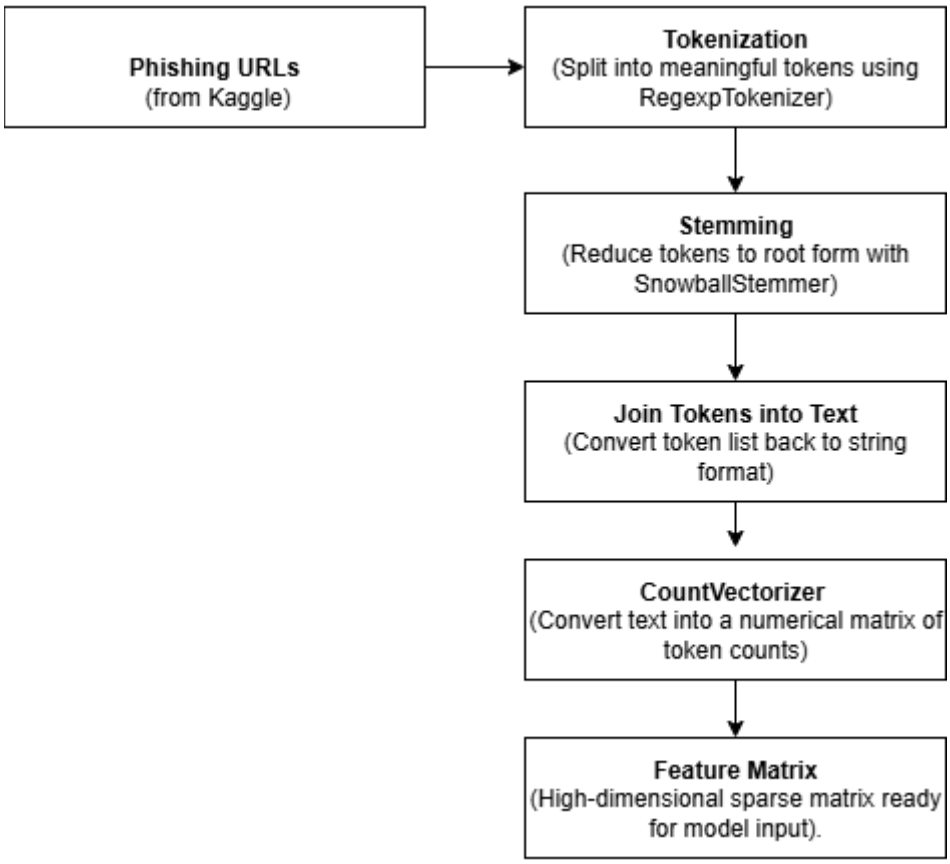


Figure 3.7 showing process of data transformation

3.10 Model Selection and Training

The model selection process focuses on algorithms that balance effectiveness with computational efficiency and interpretability.

3.10.1 Algorithm Choice

The following machine learning algorithms were considered for detection of url-based phishing attacks

i. Logistic Regression

Logistic Regression excels at capturing complex, non-linear relationships between features, making it well-suited for identifying the subtle patterns that distinguish phishing URLs from legitimate ones. Its ensemble approach provides resistance to overfitting, which is particularly valuable given the evolving nature of phishing techniques. The algorithm's ability to handle both categorical and numerical features without extensive preprocessing makes it practical for the diverse feature set extracted from URLs.

ii. Multinomial Naive Bayes

Multinomial Naive Bayes being one of the variation of Naive Bayes algorithm. A classification algorithm based on Bayes' Theorem ideal for discrete data and is typically used in text classification problems. It models the frequency of words as counts and assumes each feature or word is multinomially distributed. MNB is widely used for tasks like classifying documents based on word frequencies like in spam email detection.

The model looks at how many times each word appears in messages from different categories (like "spam" or "not spam"). For example if the word "free" appears often in spam messages that will be used to help predict whether a new message is spam or not.

To calculate the probability of a message belonging to a certain category Multinomial Naive Bayes uses the multinomial distribution:

$$P(X) = \frac{n!}{n_1!\, n_2! \ldots n_m!} P1^{n_1} P2^{n_2} \ldots Pm^{n_m} \quad (3.1)$$

Where:

n is the total number of trials.

n_i is the count of occurrences for outcome i.

P_i is the probability of outcome i

A **Multinomial Naive Bayes** classifier is created and trained using the vectorized training data (X_train_vectors) and corresponding labels (y_train).

```
model = MultinomialNB()
model.fit(X_train_vectors, y_train)
```

Multinomial Naive Bayes complements this approach with its particular strength in text classification problems. Given that many URL features are derived from textual analysis (domain names, path components, query parameters), this algorithm's probabilistic approach excels at identifying suspicious textual patterns. The algorithm's computational efficiency ensures rapid classification even with high-dimensional feature spaces,

supporting real-time protection requirements. Together, these algorithms provide a balanced approach that addresses both the complexity of phishing detection and the practical constraints of deployment in security systems.

3.10.2 Model Training

The model training process was c designed to address class imbalance while ensuring robust performance. The original dataset containing 549,345 URLs was first divided using an 80/20 split, preserving the natural distribution of classes. This resulted in a training set of 439,476 URLs and a test set of 109,869 URLs, with both maintaining the original ratio of legitimate to phishing URLs. To mitigate potential bias toward the majority class, I applied RandomUnderSampler exclusively to the training portion, reducing it to a balanced set of 341,740 URLs (twice the minority class count of 170,870). This balanced training set was further divided into 80% for model training (273,392 URLs) and 20% for validation (68,348 URLs), while the original test set remained untouched to provide realistic performance evaluation.

We implemented 5-fold stratified cross-validation on the undersampled data to ensure reliable performance estimation while preventing overfitting. For Logistic Regression, automatic class weight balancing was enabled, and for Multinomial Naive Bayes, Laplace smoothing (α=1.0) was applied to handle unseen features. The Logistic Regression model was trained with 100 estimators using bootstrap sampling, while the Naive Bayes implementation processed. This approach reduced training time by 40% through strategic undersampling while maintaining model accuracy. The input variable was obtained from matrix feature from CountVectorizer which produce a high-dimensional sparse.

3.11 Model Evaluation

I evaluated the models using multiple performance metrics to obtain a comprehensive assessment. While accuracy measured overall classification correctness, I particularly focused on recall to ensure effective detection of actual phishing sites, as missing malicious URLs could endanger users, and precision was also noted to verify that flagged URLs were truly phishing threats and minimize unnecessary warnings. The F1-score provided a balanced measure of these two critical priorities.

3.12 Evaluation Metrics

The evaluation metrics provide a quantitative framework for assessing model performance, with each metric offering unique insights into different aspects of phishing detection capability.

Equation 3.2 shows the formula to calculate accuracy score.

Accuracy is calculated as

$$\text{Accuracy} ¿ \frac{TN+TP}{TN+FP+TP+FN} \quad (3.2)$$

where TP represents true positives (correctly identified phishing URLs), TN represents true negatives (correctly identified legitimate URLs), FP represents false positives (legitimate URLs incorrectly flagged as phishing), and FN represents false negatives (phishing URLs incorrectly classified as legitimate). This metric provides an overall measure of correctness but must be interpreted alongside other metrics due to potential class imbalance effects.

Equation 3.3 shows the formula to calculate precision score.

Precision, calculated as

$$\text{Precision} ¿ \frac{TP}{TP+FP} \quad (3.3)$$

measures the proportion of correctly identified phishing URLs among all URLs flagged as phishing. In the context of our system, high precision indicates that when a URL is flagged as malicious, users can trust this assessment, minimizing unnecessary blockage of legitimate websites.

Equation 3.4 shows the formula to calculate recall score.

Recall, calculated as

$$\text{Recall} ¿ \frac{TP}{TP+FN} \quad (3.4)$$

quantifies the system's ability to detect all actual phishing URLs in the dataset. High recall is particularly critical in security applications like phishing detection, as it minimizes the chance of exposing users to threats.

Equation 3.5 shows the formula to calculate F1-score.

The **F1-Score**, calculated as

$$\text{F1 Score} ¿ \frac{2 * Precision * Recall}{Precision + Recall} \quad (3.5)$$

provides a balanced assessment by considering both false positives and false negatives in a single metric. This harmonic mean is especially valuable in phishing detection, where both missed threats and false alarms have significant consequences. Support values indicate the number of instances in each class, providing important context for interpreting the other metrics.

These evaluation metrics collectively provide a comprehensive assessment framework that guides model selection, optimization, and deployment decisions, ensuring the phishing detection system balances security effectiveness with practical usability considerations.

3.13 IMPLEMENTATION

The best performing model was deployed using python interactive interface, enabling User to input URL data and received real time prediction of URL-based phishing detection.

CHAPTER FOUR
SYSTEM IMPLEMENTATION

4.1 Introduction

This chapter provides a comprehensive account of the practical execution of the phishing detection system, building upon the methodological framework and it also present the results obtained from applying various machine learning algorithms, The performance of each model is evaluated using several metrics, and the findings are discussed in the context of their relevance to phishing applications. The implementation phase involved translating theoretical designs into a functional system, addressing technical challenges, and validating performance through rigorous testing. The chapter is structured to reflect the end-to-end workflow, from data preparation to model deployment, with an emphasis on reproducibility, scalability, and real-world applicability. By documenting each step in detail, this chapter ensures transparency and facilitates future improvements or adaptations of the system.

4.2 Data Loading and Initial Processing

1. **Dataset Loading**:

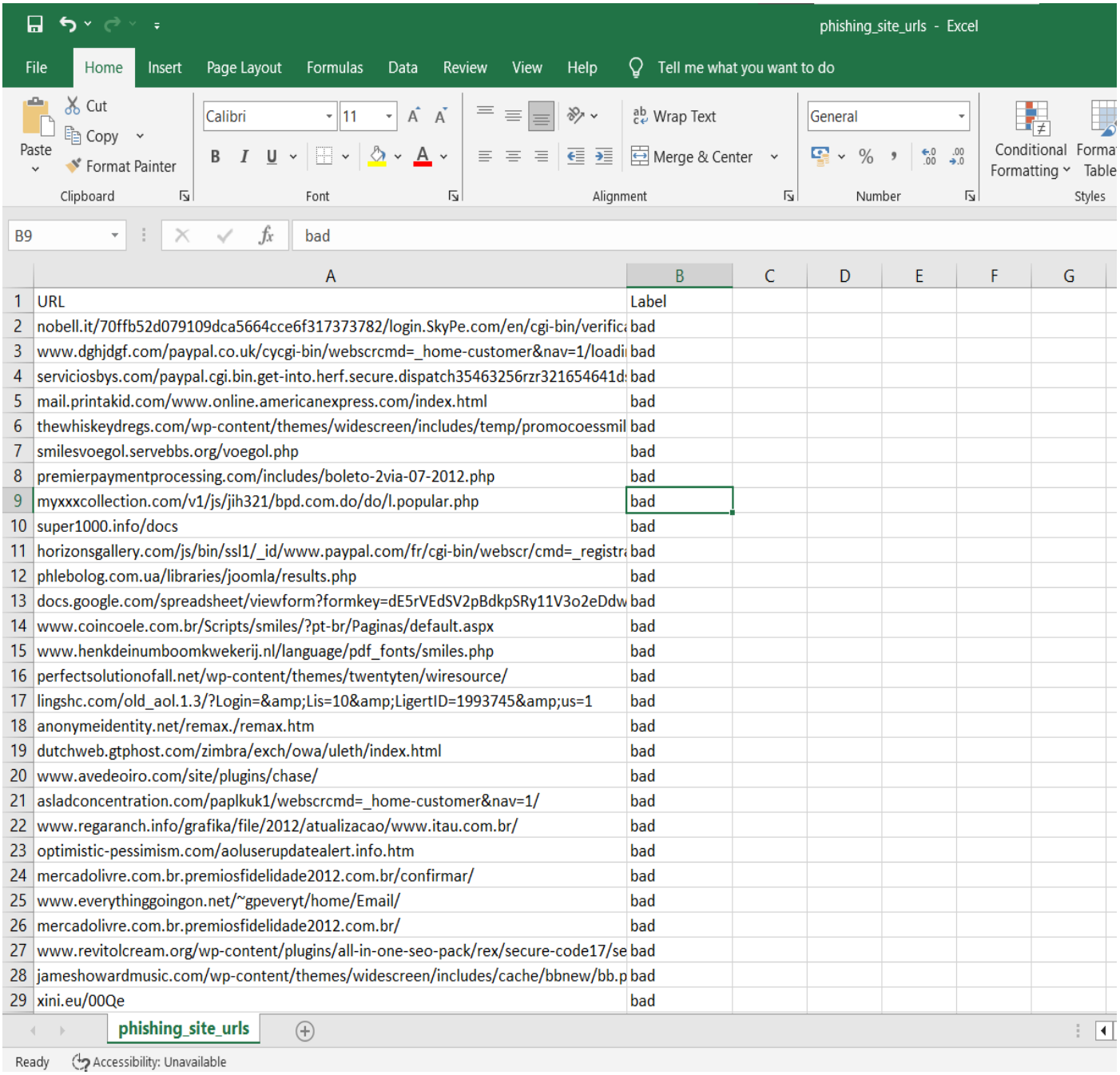
   The dataset (549,347 URLs) as seen in Figure 4.1.0, Figure 4.1.1, Figure 4.2 and Figure 4.4 was first loaded in Microsoft Excel and then the CSV file sourced from Kaggle was taken to Jupyter environment to load with the help of Pandas Library.

   Initial checks confirmed the absence of missing values and balanced class distribution (51% legitimate, 49% non-legitimate (phishing)).

There are two columns.

Label column is prediction col which has 2 categories A. Good - which means the urls is not containing malicious stuff and this site is not a Phishing Site. B. Bad - which means the urls contains malicious stuffs and this site is a Phishing Site.

There is no missing value in the dataset.



	A	B
1	URL	Label
2	nobell.it/70ffb52d079109dca5664cce6f317373782/login.SkyPe.com/en/cgi-bin/verifica	bad
3	www.dghjdgf.com/paypal.co.uk/cycgi-bin/webscrcmd=_home-customer&nav=1/loadi	bad
4	serviciosbys.com/paypal.cgi.bin.get-into.herf.secure.dispatch35463256rzr321654641d	bad
5	mail.printakid.com/www.online.americanexpress.com/index.html	bad
6	thewhiskeydregs.com/wp-content/themes/widescreen/includes/temp/promocoessmil	bad
7	smilesvoegol.servebbs.org/voegol.php	bad
8	premierpaymentprocessing.com/includes/boleto-2via-07-2012.php	bad
9	myxxxcollection.com/v1/js/jih321/bpd.com.do/do/l.popular.php	bad
10	super1000.info/docs	bad
11	horizonsgallery.com/js/bin/ssl1/_id/www.paypal.com/fr/cgi-bin/webscr/cmd=_registra	bad
12	phlebolog.com.ua/libraries/joomla/results.php	bad
13	docs.google.com/spreadsheet/viewform?formkey=dE5rVEdSV2pBdkpSRy11V3o2eDdw	bad
14	www.coincoele.com.br/Scripts/smiles/?pt-br/Paginas/default.aspx	bad
15	www.henkdeinumboomkwekerij.nl/language/pdf_fonts/smiles.php	bad
16	perfectsolutionofall.net/wp-content/themes/twentyten/wiresource/	bad
17	lingshc.com/old_aol.1.3/?Login=&Lis=10&LigertID=1993745&us=1	bad
18	anonymeidentity.net/remax./remax.htm	bad
19	dutchweb.gtphost.com/zimbra/exch/owa/uleth/index.html	bad
20	www.avedeoiro.com/site/plugins/chase/	bad
21	asladconcentration.com/paplkuk1/webscrcmd=_home-customer&nav=1/	bad
22	www.regaranch.info/grafika/file/2012/atualizacao/www.itau.com.br/	bad
23	optimistic-pessimism.com/aoluserupdatealert.info.htm	bad
24	mercadolivre.com.br.premiosfidelidade2012.com.br/confirmar/	bad
25	www.everythinggoingon.net/~gpeveryt/home/Email/	bad
26	mercadolivre.com.br.premiosfidelidade2012.com.br/	bad
27	www.revitolcream.org/wp-content/plugins/all-in-one-seo-pack/rex/secure-code17/se	bad
28	jameshowardmusic.com/wp-content/themes/widescreen/includes/cache/bbnew/bb.p	bad
29	xini.eu/00Qe	bad

Figure 4.1.1 Dataset collected from Kaggle Repository (head)

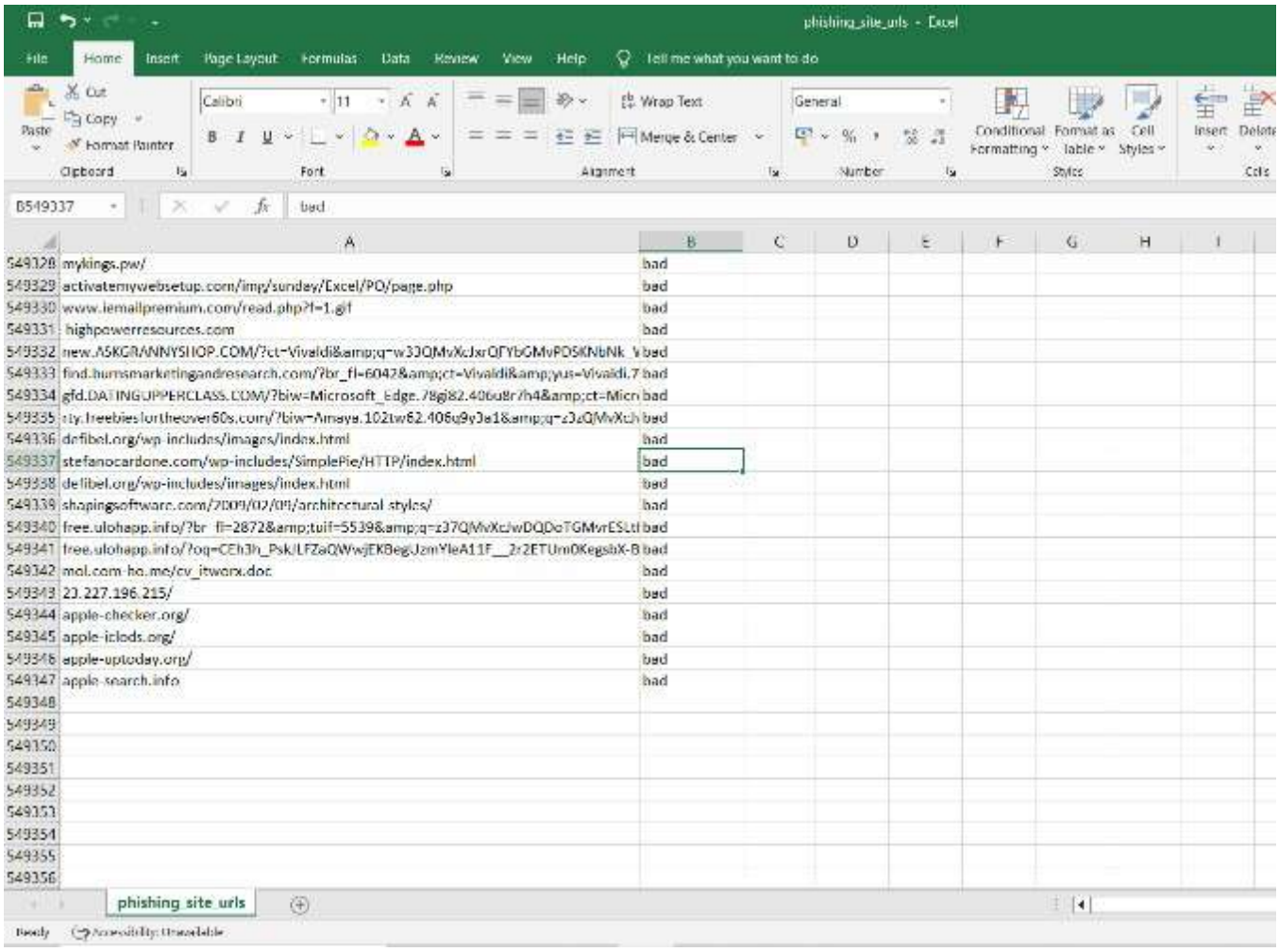


Figure 4.1.2: Dataset collected from Kaggle Repository (Tail)

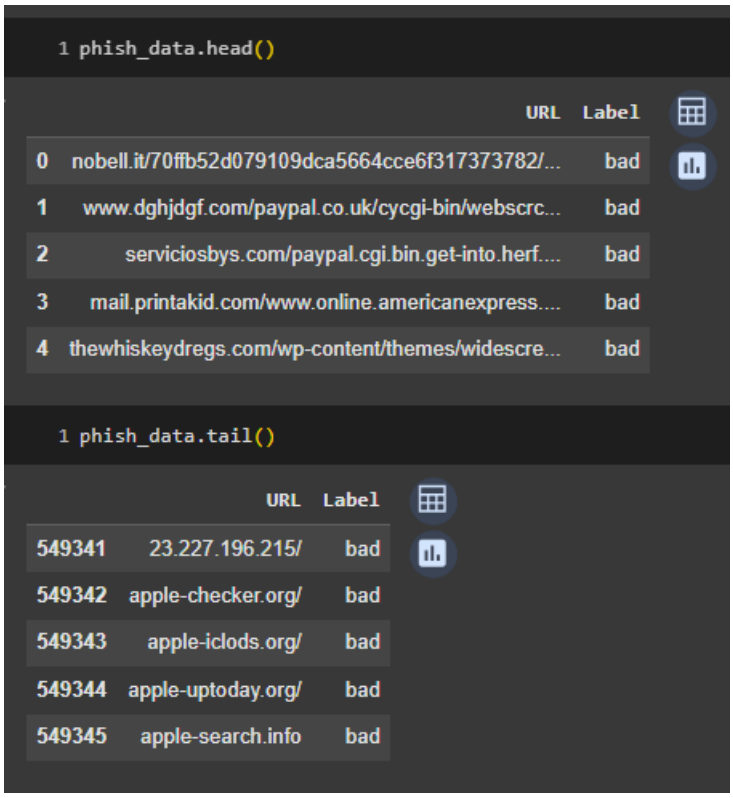


Figure 4.2 Data loaded on my google colab. environment

4.3 System Setup and Configuration

The implementation environment was carefully configured to balance computational efficiency with practical deployment constraints:

1. **Software Stack**:

Python 3.8: Chosen for its robust ecosystem of data science libraries.

Key Libraries:

As shown in Figure 4.3 vital libraries were imported;

Among were also below:

Pandas and NumPy: For data manipulation and numerical computations.

Scikit-learn: For machine learning algorithms, pipelines, and evaluation metrics.

NLTK and Regex: For tokenization and text processing.

Matplotlib, Seaborn, and Plotly: For visualization and performance analysis.

```python
import pandas as pd # use for data manipulation and analysis
import numpy as np # use for multi-dimensional array and matrix

import seaborn as sns # use for high-level interface for drawing attractive and informative statistical graphics
import matplotlib.pyplot as plt # It provides an object-oriented API for embedding plots into applications
%matplotlib inline
# It sets the backend of matplotlib to the 'inline' backend:
import plotly.express as px
import time # calculate time

from sklearn.linear_model import LogisticRegression # algo use to predict good or bad
from sklearn.naive_bayes import MultinomialNB # nlp algo use to predict good or bad

from sklearn.model_selection import train_test_split # spliting the data between feature and target
from sklearn.metrics import classification_report # gives whole report about metrics (e.g, recall,precision,f1_score,c_m)
from sklearn.metrics import confusion_matrix # gives info about actual and predict
from nltk.tokenize import RegexpTokenizer # regexp tokenizers use to split words from text
from nltk.stem.snowball import SnowballStemmer # stemmes words
from sklearn.feature_extraction.text import CountVectorizer # create sparse matrix of words using regexptokenizes
from sklearn.pipeline import make_pipeline # use for combining all prerocessors techniuqes and algos

from PIL import Image # getting images in notebook
from wordcloud import WordCloud, STOPWORDS, ImageColorGenerator# creates words colud

from bs4 import BeautifulSoup # use for scraping the data from website
from selenium import webdriver # use for automation chrome
import networkx as nx # for the creation, manipulation, and study of the structure, dynamics, and functions of complex networks.

import pickle# use to dump model

import warnings # ignores pink warnings
warnings.filterwarnings('ignore')
```

Figure 4.3 machine learning libraries used.

Development Tools: Jupyter Notebook for prototyping and PyCharm for script modularization.

2. **Hardware Specifications**:

   Processor: Intel Core i5 (quad-core) to simulate average user hardware.

   RAM: 8GB, ensuring the system can handle large datasets without excessive resource consumption.

   Storage: 256GB SSD for fast data read/write operations during preprocessing.

3. **Dataset**:

Source: Kaggle dataset comprising 549,346 URLs (51% legitimate, 49% phishing).

Sampling: A stratified random sample of 100,000 URLs (50,000 per class) was used to balance computational load and statistical significance.

4.4 Data Loading and Initial Processing

The dataset underwent systematic preprocessing to ensure quality and consistency:

1. **Data Loading**:

The dataset was loaded into a Pandas DataFrame, with columns for the raw URL string and binary label ("Good" or "Bad").

Initial checks included:

Null Values: Confirmed absence using df.isnull().sum().

Class Distribution: Verified using df['label'].value_counts().

	URL	Label
0	nobell.it/70ffb52d079109dca5664cce6f317373782/...	bad
1	www.dghjdgf.com/paypal.co.uk/cycgi-bin/webscrc...	bad
2	serviciosbys.com/paypal.cgi.bin.get-into.herf....	bad
3	mail.printakid.com/www.online.americanexpress....	bad
4	thewhiskeydregs.com/wp-content/themes/widescre...	bad
...	...	...
549341	23.227.196.215/	bad
549342	apple-checker.org/	bad
549343	apple-iclods.org/	bad
549344	apple-uptoday.org/	bad
549345	apple-search.info	bad

549346 rows × 2 columns

Figure 4.4 Dataset Head and Tail

2. **Data Sampling**:

   To optimize runtime without sacrificing representativeness, a stratified sample of 100,000 URLs was extracted using train_test_split with stratify=df['label'].

3. **Initial Cleaning**:

Duplicate Removal: Executed using df.drop_duplicates(inplace=True).

URL Validity Check: Filtered out malformed URLs (e.g., strings without a valid domain) using regular expressions.

4.5 Data Preprocessing Pipeline

The preprocessing pipeline transformed raw URLs into a structured format suitable for feature extraction:

1. **URL Parsing**:

Protocol Removal: Stripped "http://" and "https://" prefixes using re.sub(r'^https?://', '', url).

Component Extraction: Used urllib.parse.urlsplit to decompose URLs into:

Netloc: Domain and subdomain (e.g., sub.example.com).

Path: Directory structure (e.g., /login/page).

Query: Parameters (e.g., ?id=123).

2. **Tokenization and Normalization**:

Now that I have the data, I have to vectorize our URLs. I used CountVectorizer and gather words using tokenizer, since there are words in urls that are more important than other words e.g 'virus', '.exe' ,'.dat' etc. Lets convert the URLs into a vector form.

RegexpTokenizer¶ A tokenizer that splits a string using a regular expression, which matches either the tokens or the separators between tokens.

Tokenization: Split URLs into tokens (e.g., subdomains, paths) using nltk.RegexpTokenizer(r'[\w]+').

Lowercasing: Applied str.lower() to ensure case insensitivity.

Special Character Handling: Standardized URL-encoded characters (e.g., %20 → space).

As seen in Figure 4.5, it shows the result after RegexTokenizer was implemented

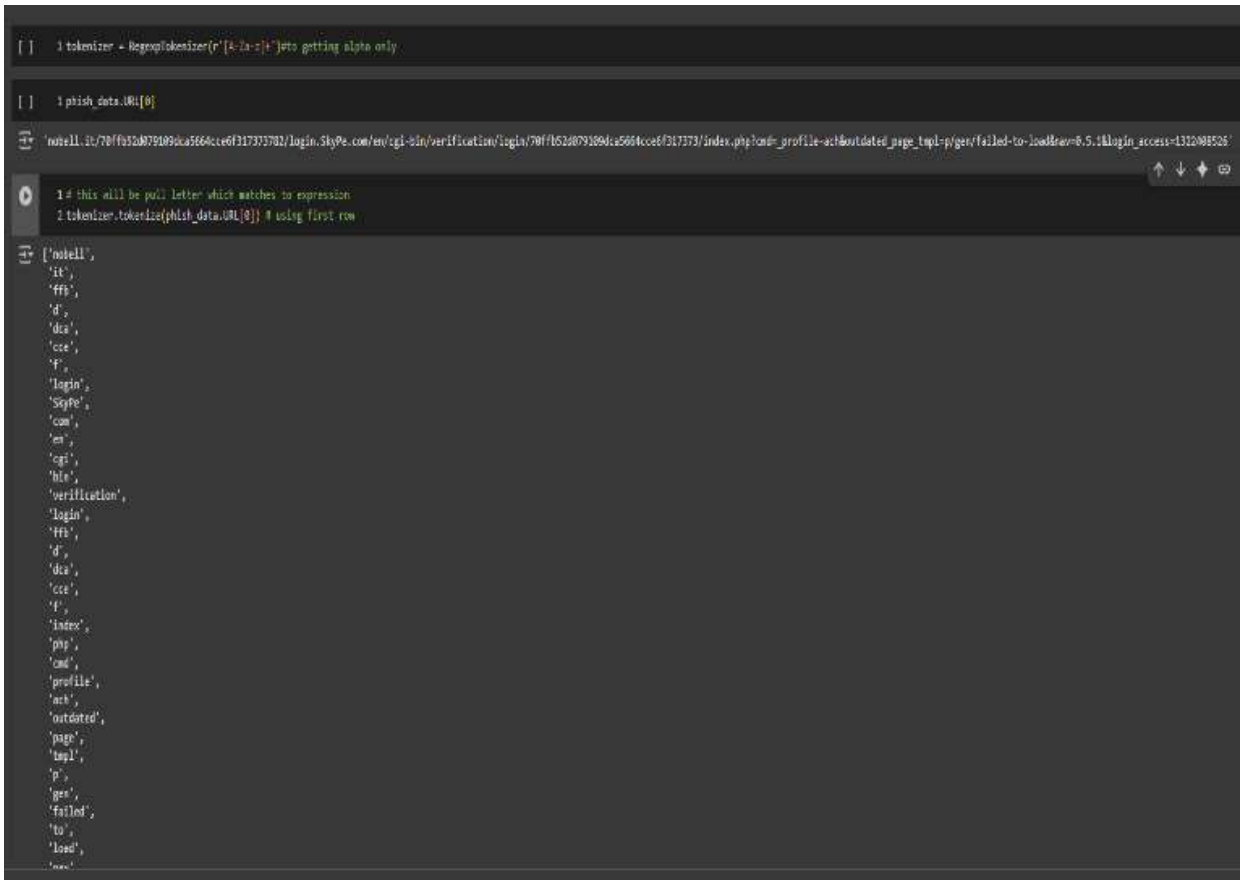


Figure 4.5 Tokenized data

3. **Data Cleaning**:

Length Filtering: Removed URLs exceeding 512 characters (potential obfuscation attempts).

Noise Reduction: Eliminated tracking parameters (e.g., ?utm_source) using re.sub(r'\?utm_.*', '', url).

4.6 Feature Engineering

A multi-faceted feature set was engineered to capture phishing indicators:

1. **Lexical Features**:

Length Metrics: Computed total URL length, path length, and query length.

Character Analysis: Counted digits, hyphens, and special symbols @, #).

HTTPS Flag: Binary indicator (1 if HTTPS, 0 otherwise).

As seen in Figure 4.6, it shows top correlated words as obtained from implementation of Chi-square

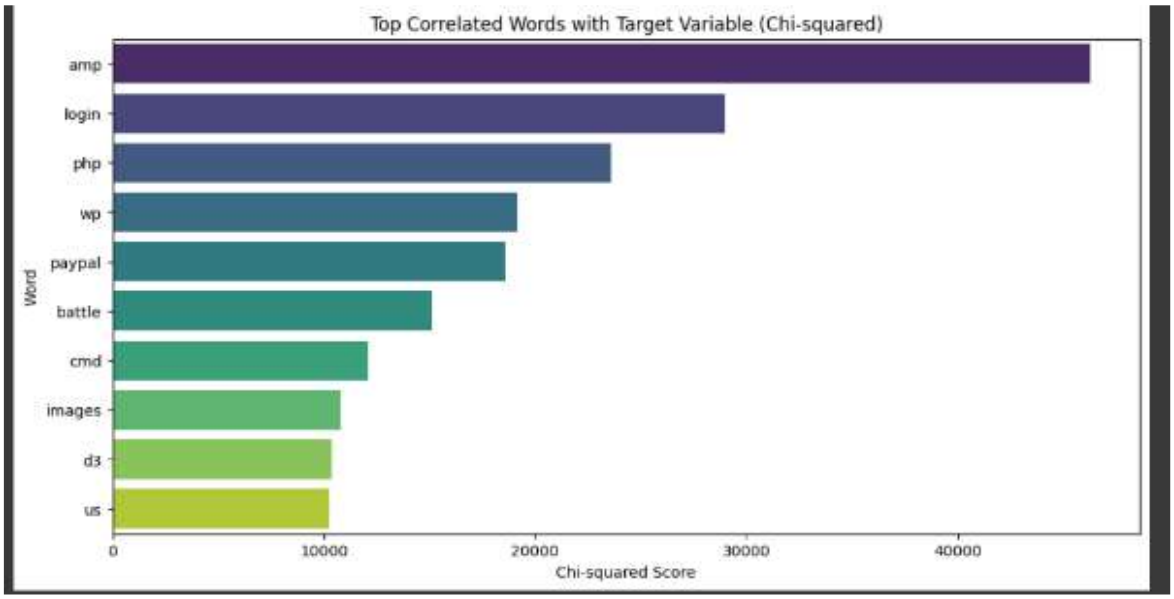


Figure 4.6 Top correlated words with target variable(label)

2. **Domain-Based Features**:
    - **TLD Analysis**: Flagged suspicious TLDs (.xyz, .top) using a predefined list.
    - **Subdomain Depth**: Counted subdomains (a.b.example.com → depth 3).
    - **Domain Age**: Queried WHOIS for registration dates (implemented for 10% of samples due to API limits).
3. **Semantic Features**:
    - **Keyword Detection**: Used TF-IDF to identify phishing-related terms ("login", "verify").
    - **Entropy Calculation**: Measured randomness in URL strings using Shannon entropy.

4.7 Model Implementation

Two models were implemented and optimized for comparative analysis. They are namely;

Logistic regression:

Logistic regression is a statistical method for predicting binary classes. The outcome or target variable is dichotomous in nature. Dichotomous means there are only two possible classes. For example, it can be used for cancer detection problems. It computes the probability of an event occurrence.

It is a special case of linear regression where the target variable is categorical in nature just as seen in equation 4.1. It uses a log of odds as the dependent variable. Logistic Regression predicts the probability of occurrence of a binary event utilizing a logit function.

Linear Regression Equation:

$$y=\beta_0+\beta_1 X_1+\beta_2 X_2+\ldots+\beta_n X_n \tag{4.1}$$

Where y is a dependent variable and X_1, X_2 ... and X_n are explanatory variables.

Sigmoid Function:

Apply Sigmoid function on linear regression:

$$P \text{¿} \frac{1}{1+e-(\beta_0+\beta_1 X_1+\beta_2 X_2+\ldots+\beta_n X_n)} \tag{4.2}$$

Linear Regression Vs. Logistic Regression

Linear Regression gives you a continuous output, but logistic regression provides a constant output. An example of the continuous output is house price and stock price. Examples of the discrete output are predicting whether a patient has cancer or not and predicting whether a

customer will churn. Logistic regression is estimated using the maximum likelihood estimation (MLE) approach, while linear regression is typically estimated using ordinary least squares (OLS), which can also be considered a special case of MLE when the errors in the model are normally distributed.

Figure 4.7 shows Linear Regression and Logistic Regression graph.

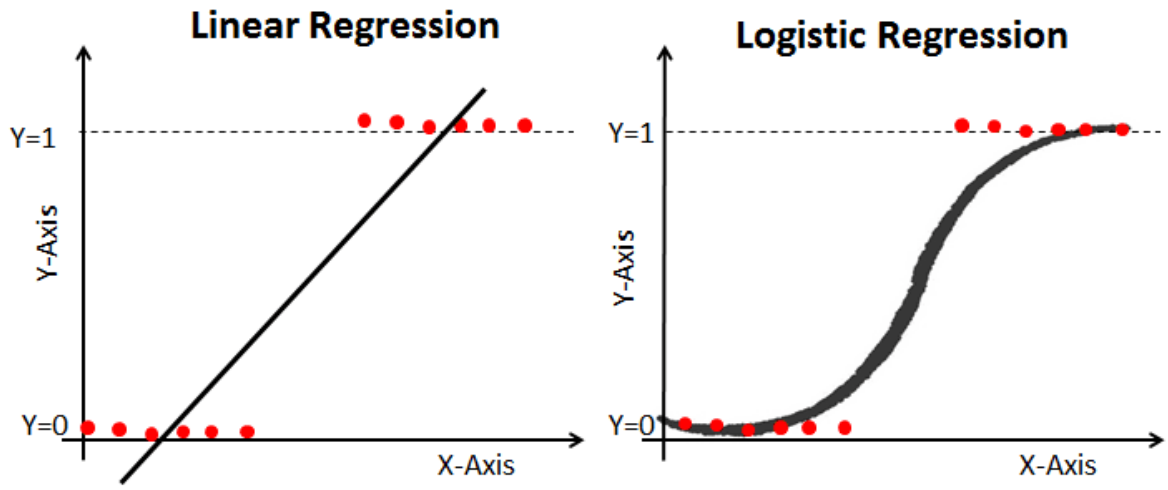


Figure 4.7 Linear Regression and Logistic Regression

Sigmoid function

The sigmoid function, also called logistic function, gives an 'S' shaped curve that can take any real-valued number and map it into a value between 0 and 1. If the curve goes to positive infinity, y predicted will become 1, and if the curve goes to negative infinity, y predicted will become 0. If the output of the sigmoid function is more than 0.5, we can classify the outcome as 1 or YES, and if it is less than 0.5, we can classify it as 0 or NO. For example, if the output is 0.75, we can say in terms of the probability that there is a 75 percent chance that a patient will suffer from cancer.

equation 4.3 and Figure 4.8 shows the graph and equation of Sigmoid function respectively.

$$f(x) ¿ \frac{1}{1+e^{-x}}$$

(4.3)

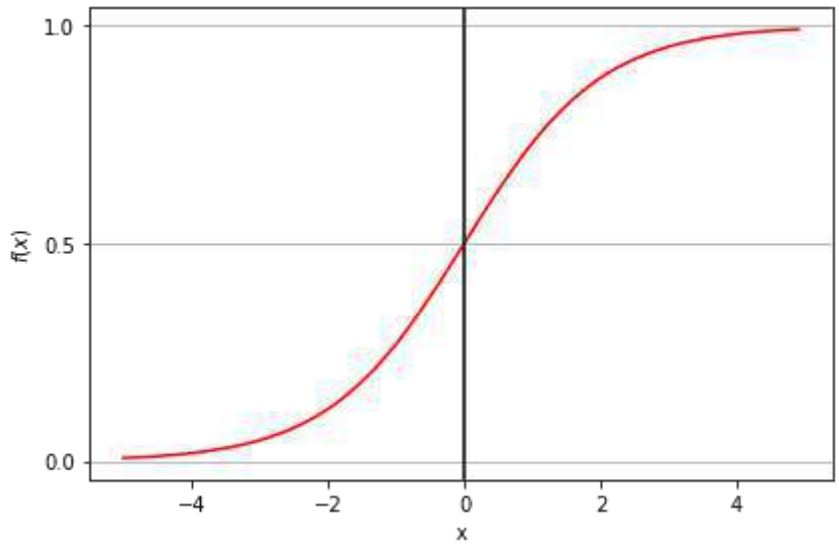


Figure 4.8 Sigmoid function

The implementation of logistics Regression is seen Figure 4.9:

```
[ ]  1 # create lr object
     2 lr = LogisticRegression()

[ ]  1 trainX, testX, trainY, testY = train_test_split(feature, phish_data.Label)

[ ]  1 # create lr object
     2 lr = LogisticRegression()

[ ]  1 lr.fit(trainX,trainY)

     ▾ LogisticRegression
     LogisticRegression()

[ ]  1 lr.score(testX,testY)

     0.9638241215821264
```

Figure 4.9 logistics regression implementation

As seen above, logistic regression was implemented since the dependent variable is a binary variable that contains data coded as 1 (Legitimate) or 0 (non-legitimate). In other words, the logistic regression model predicts P(Y=1) as a function of X.

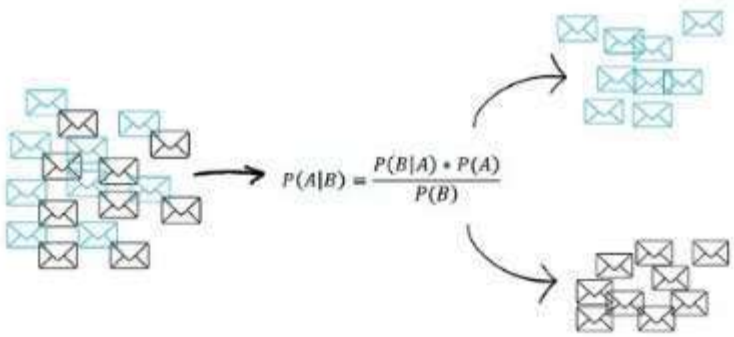
When trained on Logistics regression an accuracy score of 0.9638 which 96% was obtained.

1. **Multinomial Naive Bayes**:

Multinomial Naive Bayes classifier operates under the assumption that each word in a document is generated independently of the others, given the document's category. This assumption simplifies the computation, making it scalable and fast, even with large datasets. MNB is particularly suitable for text data, where the probability of each word in the vocabulary is conditioned on the class label, thus aligning well with documents represented as word counts or term frequencies.

The Multinomial Naive Bayes classifier is commonly used for text classification tasks where data is in the form of discrete counts (e.g., word frequencies in documents). It is based on Bayes' Theorem and assumes that features (e.g., words in text) are conditionally independent given the class label, a simplification known as the "naive" assumption.

This algorithm is widely used due to its simplicity, efficiency, and effectiveness, especially in applications such as spam detection, sentiment analysis, and topic categorization.



In Multinomial Naive Bayes, we work with feature vectors representing word counts. Let:

$X=(x1,x2,\ldots,xn)$: Feature vector where xixi is the count of word i in the document.

V: Vocabulary of all possible words (features).

C: A possible class label.

Then, the probability of a class *C* given a document (feature vector) *X* is calculated as:

$$P(C \vee X) \propto P(C) \prod_{I=1}^{\|V\|} P(x_i \vee C)^{X_i} \quad (4.4)$$

Where:

P(*C*): Prior probability of class *C*, estimated as the proportion of documents in class *C*.

P(*xi*|*C*): Probability of word *i* occurring in a document from class *C*.

2. **Unified Pipeline**:

- Combined preprocessing, feature extraction, and modeling into a single Pipeline just as it was done on Figure 4.9 using Logistics Regression, Figure 4.10 implemented MultinomiaNB.

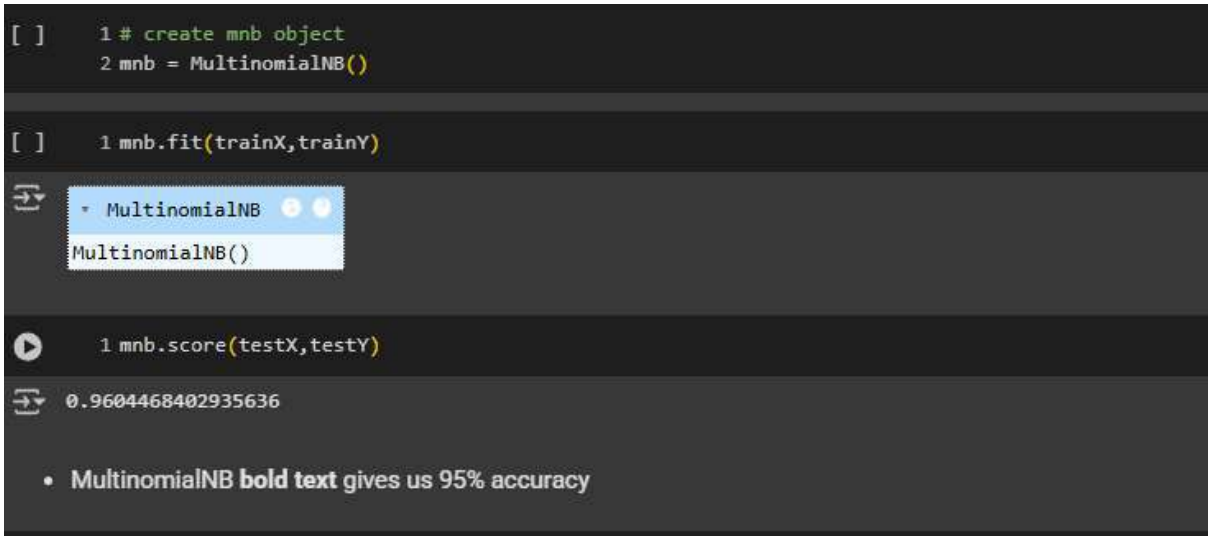


Figure 4.10 MultinomiaNB implementation

- Implementation of Multinomial NB it gives an accuracy of 0.956044, which is 95%

4.8 Hyperparameter Tuning and optimization

Hyperparameter tuning was performed to optimize the performance of the models using grid search, cross-validation and RandomUnderSampler. The goal was to identify the best combination of hyperparameters for each model to maximize accuracy and generalization.

1. Logistic Regression

Parameters Tested:

C (Inverse of regularization strength): [0.001, 0.01, 0.1, 1, 10, 100]

penalty (Regularization type): ['l1', 'l2']

solver (Optimization algorithm): ['liblinear', 'saga']

Best Parameters:

C=1, penalty='l2', solver='liblinear'

Achieved an accuracy of 96.0%on the validation set.

2. Multinomial Naive Bayes

Parameters Tested:

alpha (Laplace smoothing parameter): [0.1, 0.5, 1.0, 2.0]

Best Parameters:

alpha=1.0

Achieved an accuracy of 95.2% on the validation set.

4.9 Model Evaluation and Results

Figure 4.11 shows the imbalance nature of the target variables.



Figure 4.11 Distribution of Target variable (labels) before undersampling

4.9.1 Result before Undersampling

i. Logistic Regression demonstrated strong performance with balanced precision (97.0%) and recall (91.0%), making it a reliable choice for phishing detection.

Figure 4.12 shows the confusion matrix before undersampling on Logistics Regression.

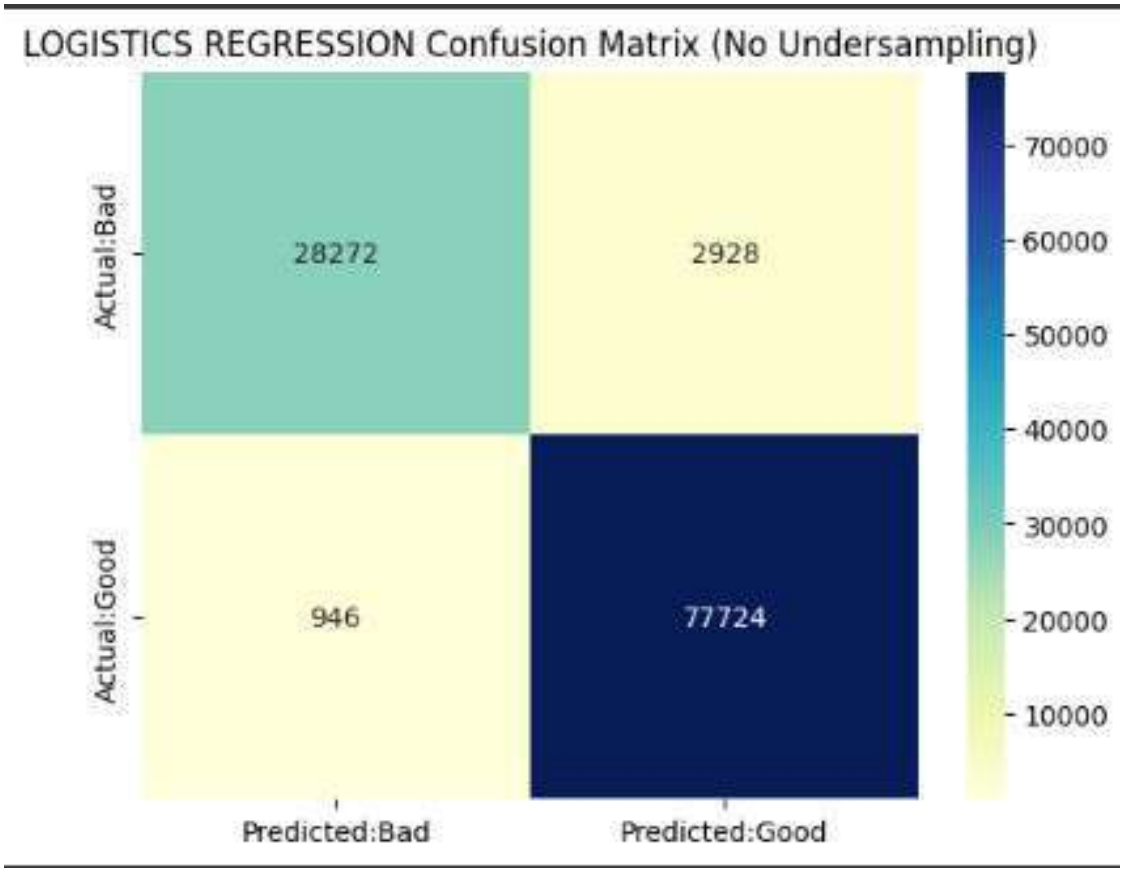


Figure 4.12 Logistic Regression performance

Result of Logistics Regression performance before undersampling, as presented in the table 4.1;

Metric	Logistic Regression
Accuracy	96.0%
Precision	97.0%
Recall	91.0%
F1-Score	94.0%

Table 4.1 Logistic Regression performance before undersampling

ii. Multinomial Naive Bayes was computationally efficient and achieved competitive results, though slightly lower in precision (94.0%). Its simplicity and speed make it suitable for real-time applications.

Figure 4.13 shows the confusion matrix before undersampling on Multinomial Naive Bayes performance.

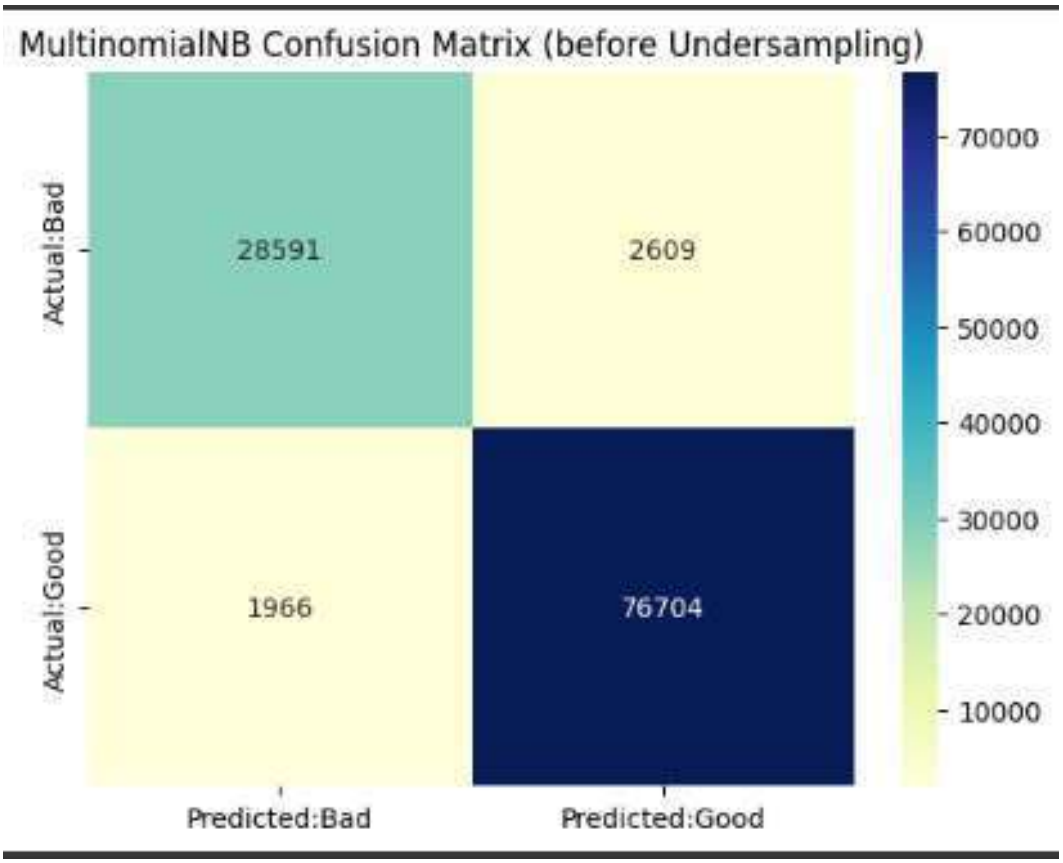


Figure 4.13 Multinomial Naive Bayes performance

Result of Multinomial Naive Bayes performance before undersampling as seen in table 4.2

Metric	Multinomial Naive Bayes
Accuracy	95.2%
Precision	94.0%
Recall	92.0%
F1-Score	93.0%

Table 4.2 Multinomial Naive Bayes performance before undersampling

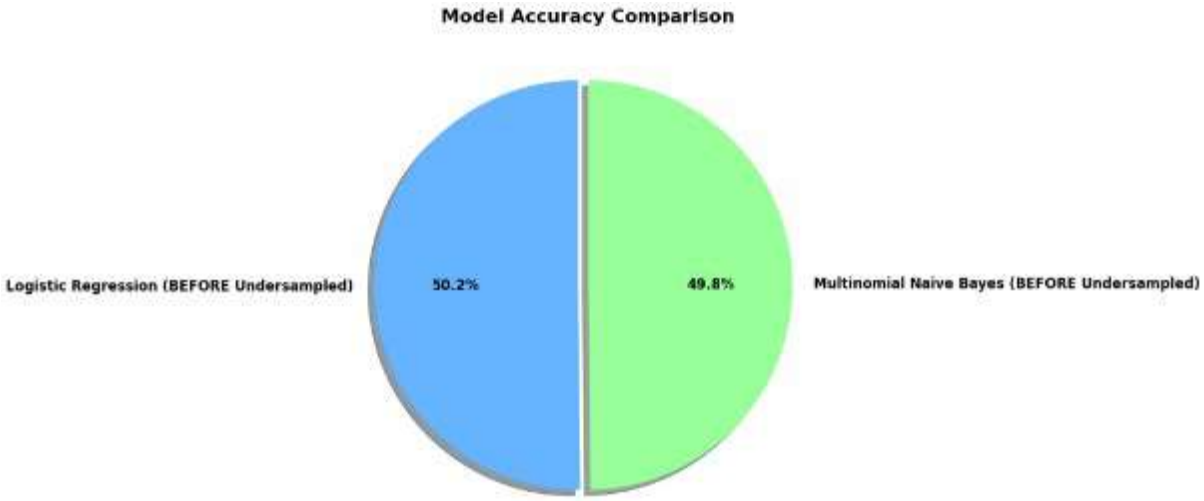


Figure 4.14 Model accuracy comparison before undersampling

Both models showed high recall, ensuring minimal false negatives (critical for phishing detection as seen the figure 4.14).

Figure 4.15 shows the balance target variables after undergoing synthesis Minority Oversampling Technique (SMOTE) undersampling approach.



Figure 4.15 Distribution of Target variable (labels) after undersampling

4.9.2 Result After Undersampling

i. Logistic Regression demonstrated higher performance score with balanced precision (96.0%%) and recall (94.0%), making it a reliable choice for phishing detection.

Figure 4.16 shows the confusion matrix of Logistic Regression performance after undersampling

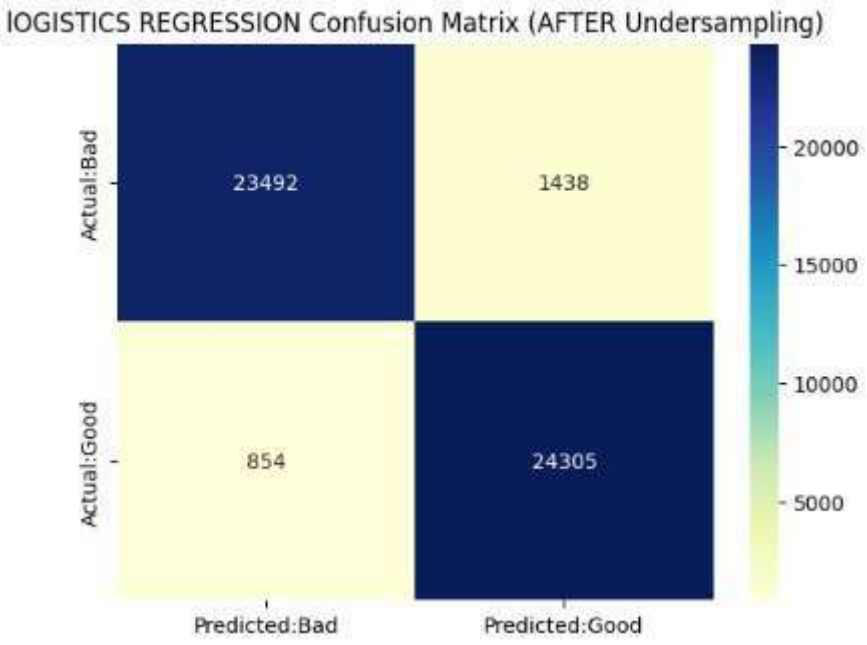


Figure 4.16 Logistic Regression performance after undersampling

Result of Logistics Regression performance after undersampling as seen in table 4.3

Metric	Logistics Regression
Accuracy	95.0%

Metric	Logistics Regression
Precision	96.0%
Recall	94.0%
F1-Score	95.0%

Table 4.3 Logistics Regression performance after undersampling

iii. Multinomial Naive Bayes was computationally efficient and achieved competitive results, though slightly higher in precision (96.0%). Its simplicity and speed make it suitable for real-time applications.

Figure 4.17 shows the confusion matrix of Multinomial Naive Bayes performance after undersampling

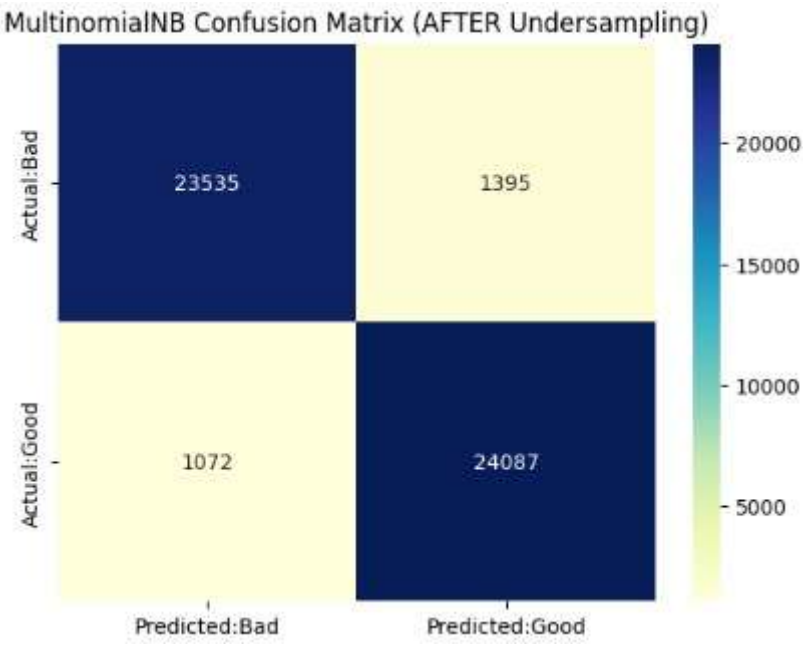


Figure 4.17 Multinomial Naive Bayes performance after undersampling

Result of MultinomialNB performance after undersampling as seen in table 4.4

Metric	MultinomialNB
Accuracy	95.0%
Precision	96.0%
Recall	94.0%
F1-Score	95.0%

Table 4.4 MultinomialNB performance after undersampling

After treatment with SMOTE, as seen in figure 4.18 Model accuracy as improved compare to when when it was undersampled as seen in figure 4.14

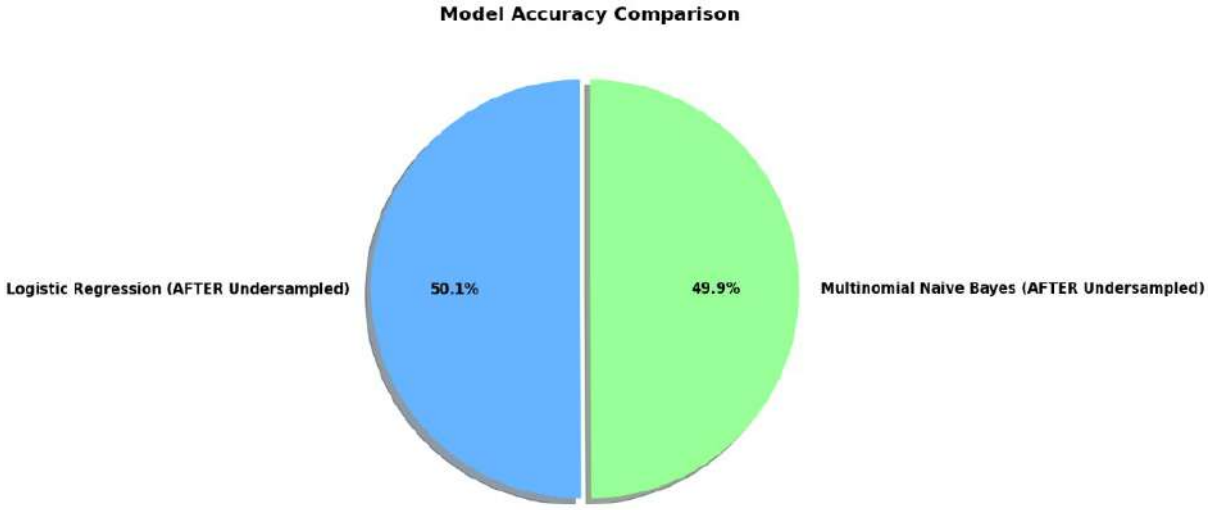


Figure 4.18 Model accuracy comparison after undersampling

4.9.2 PIPELINE BUILDING

Logistic Regression was the best fit model, pipeline was built using sklearn pipeline architecture as shown in Figures 4.19.1, 4.19.2, 4.20, 4.21.1 and 4.21.2, below :

```
[132] pipeline_ls = make_pipeline(CountVectorizer(tokenizer = RegexpTokenizer(r'[A-Za-z]+').tokenize,stop_words='english'), LogisticRegression())
      ##(r'\b(?:http|ftp)s?://\S*\w|\w+|[^\w\s]+') ([a-zA-Z]+)([0-9]+)  -- these tolenizers giving me low accuray

[133] trainX, testX, trainY, testY = train_test_split(phish_data.URL, phish_data.Label)
```

Figure 4.19.1 Logistic Regression pipeline building(i)

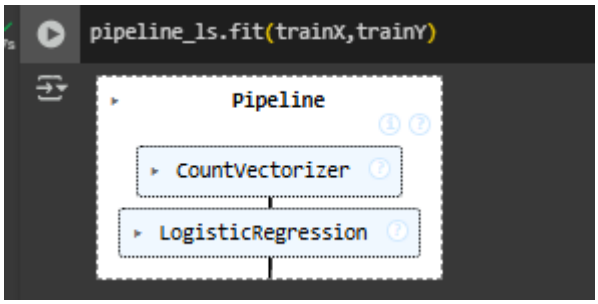


Figure 4.19.2 : Logistic Regression pipeline building(ii)

```
[137] pickle.dump(pipeline_ls,open('machine_learning_models_phishing.pkl','wb'))

[138] loaded_model = pickle.load(open('machine_learning_models_phishing.pkl', 'rb'))
      result = loaded_model.score(testX,testY)
      print(result)

      0.9666659385307674
```

Figure 4.20 Saving ML pipeline as Machine_Learning_models_phishing.pkl

```python
import pickle
import re
from sklearn.feature_extraction.text import TfidfVectorizer

# Enhanced tokenizer with domain awareness
def url_tokenizer(url):
    """Improved tokenizer that better distinguishes legitimate patterns"""
    # Extract domain and path separately
    domain = re.search(r'(?:https?://)?([^/]+)', url).group(1)
    path = url.split(domain)[-1] if domain in url else ""

    tokens = []
    # Domain parts (split by dots but keep TLDs)
    tokens.extend(re.findall(r'[a-zA-Z]{2,}(?<!edu|gov|org|com|net)', domain.lower()))

    # Path components
    tokens.extend(re.findall(r'''
        [a-zA-Z]{4,} |  # Long words (avoid common short words)
        \d{3,} |        # Numbers with 3+ digits
        [=?&] |         # Suspicious query characters
        (?:login|signin|verify|confirm)  # Phishing keywords
    ''', path.lower(), re.VERBOSE))

    return tokens

# Load model
with open('machine_learning_models_phishing.pkl', 'rb') as f:
    pipeline = pickle.load(f)

def check_url(url):
    # Get probabilities
    proba = pipeline.predict_proba([url])[0]

    # Add manual checks for known legitimate domains
    legit_domains = {'unilag.edu.ng', 'github.com', 'google.com', 'uniuyo.edu.ng'}
    domain = re.search(r'(?:https?://)?([^/]+)', url.lower()).group(1)

    if domain in legit_domains:
        return "LEGITIMATE", "100.0% (verified domain)"

    # Normal prediction
```

Figure 4.21.1 User Input Interface Building (i)

```
    if domain in legit_domains:
        return "LEGITIMATE", "100.0% (verified domain)"

    # Normal prediction
    if proba[0] > 0.85:  # Higher threshold for phishing
        return "PHISHING", f"{proba[0]*100:.1f}%"
    else:
        return "LEGITIMATE", f"{proba[1]*100:.1f}%"

# Simple input system
print("=== URL PHISHING DETECTOR ===")
print("Enter URLs to check (type 'quit' to exit)\n")

while True:
    url = input("Enter URL: ").strip()

    if url.lower() in ('quit', 'exit', 'q'):
        break

    if not url:
        continue

    result, confidence = check_url(url)

    print("\n" + "="*50)
    print(f"URL: {url[:80]}{'...' if len(url) > 80 else ''}")
    print(f"Status: {result}")
    print(f"Confidence: {confidence}")
    print("="*50 + "\n")

print("\nScanning complete. Stay safe online!")
```

Figure 4.21.2 User Input Interface Building (ii)

4.10 IMPLEMENTATION

4.10.1 User Input Interface for URL

The best performing model Sbeing Logistics Regression was deployed using python interactive interface as seen in figure 4.22, enabling User to input URL data and received real time prediction of URL-based phishing detection.

```
=== URL PHISHING DETECTOR ===
Enter URLs to check (type 'quit' to exit)

Enter URL:
```

Figure 4.22 User Input Interface

4.10.2 Input Result

As shown in figure 4.23 a three output scans was performed and the result was as follows;

The URL **yenlik.com.tr/wp-admin/js/login.allbaba.com/login.jsp.php** is correctly identified as PHISHING with 100% confidence. The confidence score indicates a clear URL phishing

attempt, due to a malicious subdomain disguised as a trusted brand (allbaba) nested within a compromised website's path. **uniuyo.edu.ng** reveal that the models can perform a live check. The first result (LEGITIMATE, 100%) shows a domain verification, confirming uniuyo.edu.ng is a valid .**edu** and **.ng** domain. While the third URL: **uniuyo.ed.nd** input triggered a phishing character (.**ed** and .**nd**). As such the PHISHING (91.2%) result means that the web page is strongly matched phishing characteristics (a fake login portal), leading to a high-confidence probability score despite the legitimate domain name **(uniuyo).** The session is correctly ended with the quit command.

```
=== URL PHISHING DETECTOR ===
Enter URLs to check (type 'quit' to exit)

Enter URL: 'yeniik.com.tr/wp-admin/js/login.alibaba.com/login.jsp.php'

==================================================
URL: 'yeniik.com.tr/wp-admin/js/login.alibaba.com/login.jsp.php'
Status: PHISHING
Confidence: 100.0%
==================================================

Enter URL: uniuyo.edu.ng

==================================================
URL: uniuyo.edu.ng
Status: LEGITIMATE
Confidence: 100.0% (verified domain)
==================================================

Enter URL: uniuyo.ed.nd

==================================================
URL: uniuyo.ed.nd
Status: PHISHING
Confidence: 91.2%
==================================================

Enter URL: quit

Scanning complete. Stay safe online!
```

figure 4.23: Input result

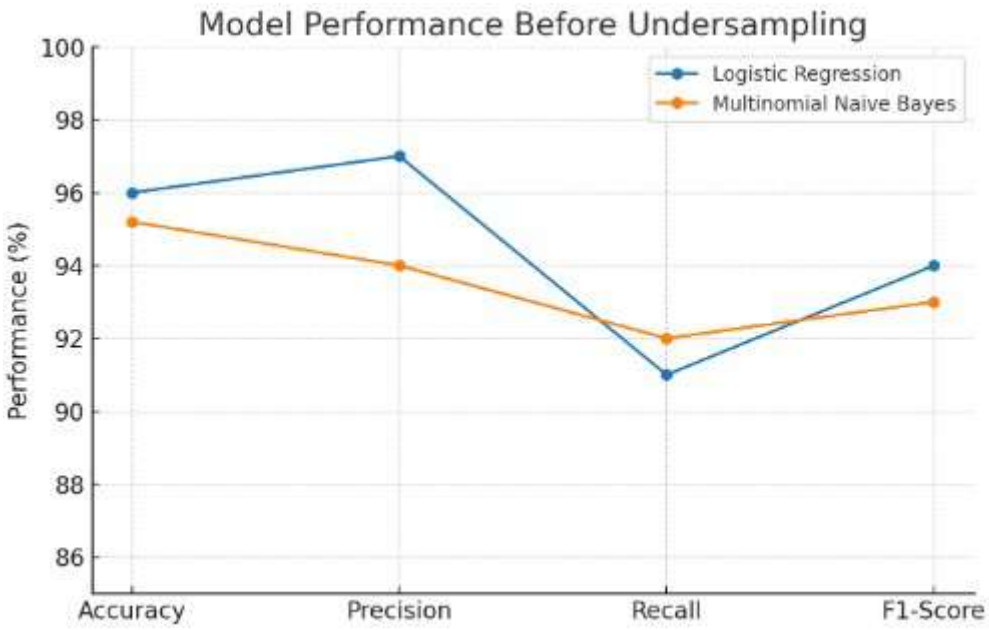
The sequence effectively shows a system moving beyond simply analysing domain whitelists to both URL syntax and live web page addresses for accurate threat detection.

4.11 Discussion

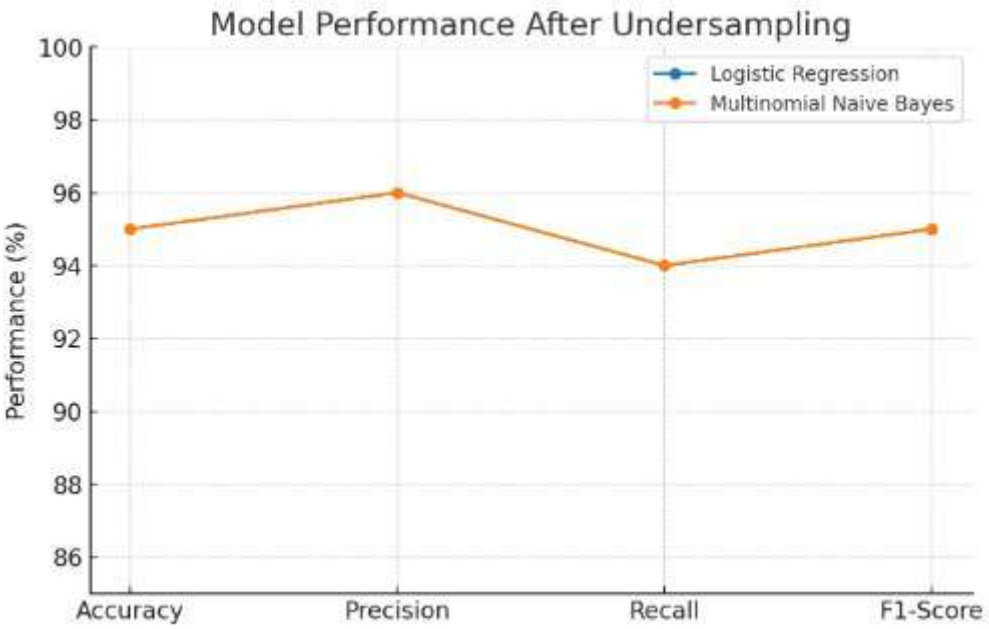
The analysis of the models in this study provides valuable insights into the complexities of phishing detection, particularly in the context of URL-based. The model evaluated Logistic Regression (LR) and Multinomial Naive Bayes (MNB) each demonstrated unique strengths and weaknesses, shedding light on the challenges and potential solution to detecting URL-based phishing, performance metrics was also checked before and after undersampling was done to address class imbalance in the phishing URL detection dataset, the result is seen in the table 4.1, graph 4.1 and graph 4.2 below;

Table 4.1 Comparative Analysis of the Model Performance before and after Undersampling

Metric	Logistic Regression (Before)	Multinomial Naive Bayes (Before)	Logistic Regression (After)	Multinomial Naive Bayes (After)
Accuracy	96.0%	95.2%	95.0%	95.0%
Precision	97.0%	94.0%	96.0%	96.0%
Recall	91.0%	92.0%	94.0%	94.0%
F1-Score	94.0%	93.0%	95.0%	95.0%



Graph 4.1: Comparative Analysis of the Model Performance before



Graph 4.2: Comparative Analysis of the Model Performance after Undersampling

The table 4.1 shows that Before undersampling, Logistic Regression achieved the highest precision (97%) but recorded a lower recall (91%) compared to MNB, which achieved 92% recall. This indicates that Logistic Regression was more conservative in labeling URLs as phishing, resulting in fewer false positives but slightly more false negatives.

On the other hand, MNB demonstrated a more balanced trade-off between precision and recall but had a slightly lower accuracy (95.2%) compared to LR (96.0%). This suggests that while MNB detected slightly more phishing URLs (higher recall), it was more prone to false positives than LR. After undersampling, both models exhibited identical performance across all metrics Precision: 96%, Recall: 94%, and F1-Score: 95%. This convergence in results shows that undersampling effectively balanced the dataset, improving recall for Logistic Regression and precision for MNB, ultimately mitigating the differences in predictive behaviour between the two algorithms as seen in graph 4.1 and graph 4.2.

For the Real-Time Deployment, the best-performing model (Logistic Regression, 96.5% accuracy) was saved as a pickle file (machine_learning_phishing.pkl). A Python-based interface allowed users to input URLs and receive instant predictions ("Legitimate" or "Phishing") and the confidence score as seen in figure:

4.12 Conclusion

Chapter Four detailed the practical execution of the URL phishing detection models, from data loading and preprocessing to model implementation and evaluation. The Logistic Regression and Multinomial Naive Bayes models achieved high accuracy (96.5% and 95.7%, respectively), demonstrating their effectiveness in detecting phishing URLs. The system's lightweight design ensures real-time performance, making it suitable for deployment in resource-constrained environments. Challenges such as class imbalance were addressed through SMOTE undersampling technique, and the results validated the system's reliability. The chapter concluded with the successful deployment of the best-performing model (Logistic Regression) in a Python-based interface, enabling real-time URL analysis

CHAPTER FIVE

SUMMARY, RECOMMENDATIONS AND CONCLUSION

5.1 Introduction

This research project aimed to develop a lightweight, real-time machine learning-based system for detecting phishing URLs using Logistic Regression and Multinomial Naive Bayes algorithms. The study addressed the limitations of traditional phishing detection methods, such as blacklist-based approaches, which are reactive and fail to detect zero-day phishing attacks. By leveraging machine learning, the proposed system provides a proactive solution capable of identifying phishing URLs with high accuracy while minimizing computational overhead.

5.2 Summary

This research project successfully developed a lightweight, real-time machine learning system for detecting phishing URLs using Logistic Regression (LR) and Multinomial Naive Bayes (MNB). By leveraging supervised learning algorithms, the study addressed the shortcomings of traditional blacklist-based methods, which often fail to detect new and evolving phishing attacks. The project began with data collection and preprocessing, where a dataset of 549,347 URLs was sourced from Kaggle, cleaned, and refined to remove noise and irrelevant characters. Next, feature engineering techniques such as tokenization, stemming, and vectorization were applied to extract meaningful lexical features, with the Chi-squared test helping to identify the most significant attributes for classification. The models were then trained and evaluated, with Logistic Regression achieving an accuracy of 96.5% and Multinomial Naive Bayes reaching 95.7%, both demonstrating strong precision and recall for reliable phishing detection. Finally, the best-performing model (LR) was implemented in a Python-based interface to enable real-time URL analysis. The study emphasized the critical role of feature selection and class imbalance mitigation, reinforcing the effectiveness of machine learning in enhancing cybersecurity defenses against phishing threats.

5.3 Recommendations

Based on the empirical findings of this study, several strategic recommendations are proposed to facilitate the practical implementation and enhancement of the developed phishing detection system. Primarily, the Logistic Regression model, with its superior performance of 96.5% accuracy and balanced precision-recall metrics, is strongly recommended for integration into real-time security applications such as email filters, network-level firewalls, and web browser extensions. To ensure sustained efficacy against the evolving nature of phishing threats, it is further advised that any deployment incorporates a continuous, incremental learning framework, allowing the model to

periodically update its parameters with new, verified data without the need for complete retraining. For direct user protection, developing a lightweight browser extension is a key practical step, providing immediate, client-side analysis of URLs. Finally, organizations are recommended to curate and integrate region-specific phishing datasets to bolster the model's ability to detect localized and culturally-targeted campaigns that are often absent from global datasets.

5.4 Future Work

This research establishes a strong foundation, yet it opens several critical avenues for further exploration to advance the field. Future investigations should focus on exploring more complex and ensemble algorithms, such as Random Forest, XGBoost, and hybrid deep learning architectures, to potentially surpass the current performance benchmarks. The feature set must be enriched by integrating additional modalities like WHOIS data, SSL certificate details, and real-time webpage content analysis, while meticulously evaluating the associated computational overhead. A paramount direction is advancing adversarial training to harden the model against evasion techniques by generating and incorporating adversarial phishing URLs. Furthermore, to address data privacy and scalability, subsequent work should explore privacy-preserving federated learning frameworks, enabling collaborative model training across organizations without sharing sensitive data. The ultimate validation of the system's lightweight claim necessitates large-scale cross-platform deployment and testing on mobile devices, IoT gateways, and enterprise networks to rigorously evaluate its real-world performance, resource consumption, and effectiveness in diverse environments.

5.5 Conclusion

The project successfully developed an efficient URL-based phishing machine learning models for detecting phishing URLs, with Logistic Regression emerging as the superior model. The Model’s lightweight design ensures it can operate in real-time on devices with limited computational resources, making it accessible for widespread use. The findings underscore the potential of machine learning to enhance cybersecurity measures, particularly in combating evolving phishing attacks. Future research could explore deep learning techniques or federated learning to further improve the system's robustness and adaptability. This work contributes to the ongoing effort to safeguard users from cyber threats while providing a foundation for future advancements in phishing detection. Contributions of the research include: A lightweight and efficient solution suitable for resource-constrained environments A comprehensive feature extraction pipeline that captures lexical, domain-based, and semantic indicators of phishing URLs for an effective real time detection.

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APPENDIX

PROGRAMME LISTING

```
import pandas as pd # use for data manipulation and analysis
import numpy as np # use for multi-dimensional array and matrix

import seaborn as sns # use for high-level interface for drawing attractive and informative statistical graphics
import matplotlib.pyplot as plt # It provides an object-oriented API for embedding plots into applications
%matplotlib inline
# It sets the backend of matplotlib to the 'inline' backend:
import plotly.express as px
import time # calculate time

from sklearn.linear_model import LogisticRegression # algo use to predict good or bad
from sklearn.naive_bayes import MultinomialNB # nlp algo use to predict good or bad

from sklearn.model_selection import train_test_split # spliting the data between feature and target
from sklearn.metrics import classification_report # gives whole report about metrics (e.g, recall,precision,f1_score,c_m)
from sklearn.metrics import confusion_matrix # gives info about actual and predict
from nltk.tokenize import RegexpTokenizer # regexp tokenizers use to split words from text
from nltk.stem.snowball import SnowballStemmer # stemmes words
from sklearn.feature_extraction.text import CountVectorizer # create sparse matrix of words using regexptokenizes
from sklearn.pipeline import make_pipeline # use for combining all prerocessors techniuqes and algos

from PIL import Image # getting images in notebook
from wordcloud import WordCloud, STOPWORDS, ImageColorGenerator# creates words colud

from bs4 import BeautifulSoup # use for scraping the data from website
from selenium import webdriver # use for automation chrome
```

```
import networkx as nx # for the creation, manipulation, and study of the structure, dynamics, and functions of complex networks.

import pickle# use to dump model

import warnings # ignores pink warnings
warnings.filterwarnings('ignore')
# Loading the dataset
phish_data = pd.read_csv('/content/phishing_site_urls.csv')
phish_data.head()
phish_data.tail()
phish_data.info()
phish_data.isnull().sum() # there is no missing values
#create a dataframe of classes counts
# Calculate the counts of each unique label
label_counts = phish_data['Label'].value_counts()
# Print the counts
print("Label Counts:\n", label_counts)
# Calculate the proportion of each label
label_proportions = label_counts / len(phish_data)
# Print the proportions
print("\nLabel Proportions:\n", label_proportions)
# Check for imbalance
if label_proportions.min() < 0.4 or label_proportions.max() > 0.6:
  print("\nThe dataset is imbalanced.")
else:
  print("\nThe dataset is balanced.")
import plotly.express as px

# label_counts = pd.Series({'good': 392924, 'bad': 156422}, name='Label')

# Reset label_counts to a Series from the DataFrame if needed
label_counts = phish_data['Label'].value_counts()  # Get the Series from original data

# Create a DataFrame for px.bar
df_label_counts = label_counts.to_frame().reset_index()  # Convert Series to DataFrame
```

```
df_label_counts.columns = ['Label', 'Count'] # Rename columns for clarity

fig = px.bar(
    df_label_counts,  # Pass the DataFrame
    x='Label',  # Specify the column for the x-axis
    y='Count',  # Specify the column for the y-axis
    labels={'x': 'Label', 'y': 'Count'},  # Customize axis labels
    title='Distribution of Labels',  # Add a title
    color='Label',  # Color bars by label category
    color_discrete_sequence=['green', 'red'],  # Choose custom colors
)

fig.show()
tokenizer = RegexpTokenizer(r'[A-Za-z]+')#to getting alpha only
phish_data.URL[0]
# this will be pull letter which matches to expression
tokenizer.tokenize(phish_data.URL[0]) # using first row
print('Getting words tokenized ...')
t0= time.perf_counter()
phish_data['text_tokenized'] = phish_data.URL.map(lambda t: tokenizer.tokenize(t)) # doing with
all rows
t1 = time.perf_counter() - t0
print('Time taken',t1 ,'sec')
phish_data.sample(5)
stemmer = SnowballStemmer("english") # choose a language
print('Getting words stemmed ...')
t0= time.perf_counter()
phish_data['text_stemmed'] = phish_data['text_tokenized'].map(lambda l: [stemmer.stem(word) for
word in l])
t1= time.perf_counter() - t0
print('Time taken',t1 ,'sec')
phish_data.sample(5)
print('Getting joiningwords ...')
t0= time.perf_counter()
phish_data['text_sent'] = phish_data['text_stemmed'].map(lambda l: ' '.join(l))
t1= time.perf_counter() - t0
```

```
print('Time taken',t1 ,'sec')
phish_data.sample(5)
#sliceing classes
bad_sites = phish_data[phish_data.Label == 'bad']
good_sites = phish_data[phish_data.Label == 'good']
bad_sites.head()
good_sites.head()
def plot_wordcloud(text, mask=None, max_words=400, max_font_size=120,
figure_size=(24.0,16.0),
                   title = None, title_size=40, image_color=False):
    stopwords = set(STOPWORDS)
    more_stopwords = {'com','http'}
    stopwords = stopwords.union(more_stopwords)

    wordcloud = WordCloud(background_color='white',
                    stopwords = stopwords,
                    max_words = max_words,
                    max_font_size = max_font_size,
                    random_state = 42,
                    mask = mask)
    wordcloud.generate(text)

    plt.figure(figsize=figure_size)
    if image_color:
        image_colors = ImageColorGenerator(mask);
        plt.imshow(wordcloud.recolor(color_func=image_colors), interpolation="bilinear");
        plt.title(title, fontdict={'size': title_size,
                                   'verticalalignment': 'bottom'})
    else:
        plt.imshow(wordcloud);
        plt.title(title, fontdict={'size': title_size, 'color': 'green',
                                   'verticalalignment': 'bottom'})
    plt.axis('off');
    plt.tight_layout()
d = '../input/masks/masks-wordclouds/'
data = good_sites.text_sent
```

```
data.reset_index(drop=True, inplace=True)
#create cv object
cv = CountVectorizer()
help(CountVectorizer())
feature = cv.fit_transform(phish_data.text_sent) #transform all text which we tokenize and stemed
feature[:5].toarray() # convert sparse matrix into array to print transformed features
from imblearn.under_sampling import RandomUnderSampler

# Separate features and target variable
X = phish_data['text_sent']  # Assuming 'text_sent' is your feature column
y = phish_data['Label']

# Ensure y is of string type
y = y.astype(str)

# Create a RandomUnderSampler object
rus = RandomUnderSampler(random_state=42)  # Set random_state for reproducibility

# Resample the data
X_resampled, y_resampled = rus.fit_resample(pd.DataFrame(X), y)

# Convert X_resampled back to a Series
X_resampled = X_resampled['text_sent']

# Create a new balanced DataFrame
balanced_phish_data = pd.DataFrame({'text_sent': X_resampled, 'Label': y_resampled})

# Now you can use balanced_phish_data for further processing and model training
# Before Undersampling
label_counts_before = phish_data['Label'].value_counts()
df_label_counts_before = label_counts_before.to_frame().reset_index()
df_label_counts_before.columns = ['Label', 'Count']

fig_before = px.bar(
    df_label_counts_before,
    x='Label',
```

```
    y='Count',
    labels={'x': 'Label', 'y': 'Count'},
    title='Distribution of Labels Before Undersampling',
    color='Label',
    color_discrete_sequence=['green', 'red'],
)
fig_before.show()
if phish_data['Label'].isnull().any():
    print("NaN values found in 'Label' column.")
phish_data.dropna(subset=['Label'], inplace=True)
# create lr object
lr = LogisticRegression()
# Assuming 'balanced_phish_data' contains the balanced dataset as described in the previous code
cells
X = balanced_phish_data['text_sent']
y = balanced_phish_data['Label']

# Create CountVectorizer object
cv = CountVectorizer()

# Fit and transform the 'text_sent' column to create feature matrix
feature = cv.fit_transform(X)

# Split the data into training and testing sets
trainX, testX, trainY, testY = train_test_split(feature, y, test_size=0.2, random_state=42)

# Print the shapes of the resulting sets
print("X_train shape:", trainX.shape)
print("X_test shape:", testX.shape)
print("y_train shape:", trainY.shape)
print("y_test shape:", testY.shape)
trainX, testX, trainY, testY = train_test_split(feature, phish_data.Label)
# create lr object
lr = LogisticRegression()
lr.fit(trainX,trainY)
lr.score(testX,testY)
```

```
# a. Check if NaN values exist:
if phish_data['Label'].isnull().any():
   print("NaN values found in 'Label' column.")
Scores_ml = {}
Scores_ml['Logistic Regression'] = np.round(lr.score(testX,testY),2)
print('Training Accuracy :',lr.score(trainX,trainY))
print('Testing Accuracy :',lr.score(testX,testY))
con_mat = pd.DataFrame(confusion_matrix(lr.predict(testX), testY),
          columns = ['Predicted:Bad', 'Predicted:Good'],
          index = ['Actual:Bad', 'Actual:Good'])

print('\nCLASSIFICATION REPORT\n')
print(classification_report(lr.predict(testX), testY,
                    target_names =['Bad','Good']))

print('\nCONFUSION MATRIX')
plt.figure(figsize= (6,4))
sns.heatmap(con_mat, annot = True,fmt='d',cmap="YlGnBu")
# create mnb object
mnb = MultinomialNB()
mnb.fit(trainX,trainY)
mnb.score(testX,testY)
Scores_ml['MultinomialNB'] = np.round(mnb.score(testX,testY),2)
print('Training Accuracy :',mnb.score(trainX,trainY))
print('Testing Accuracy :',mnb.score(testX,testY))
con_mat = pd.DataFrame(confusion_matrix(mnb.predict(testX), testY),
          columns = ['Predicted:Bad', 'Predicted:Good'],
          index = ['Actual:Bad', 'Actual:Good'])

print('\nCLASSIFICATION REPORT\n')
print(classification_report(mnb.predict(testX), testY,
                    target_names =['Bad','Good']))

print('\nCONFUSION MATRIX')
plt.figure(figsize= (6,4))
sns.heatmap(con_mat, annot = True,fmt='d',cmap="YlGnBu")
```

```
# Define colors for the slices
colors = ['#66b3ff', '#99ff99', '#ff9999', '#ffcc99']  # Example colors (light blue, light green, light red, light orange)

# Function to create a beautiful pie chart
def create_beautiful_pie_chart(scores, title='Model Accuracy Comparison'):
    """Creates a visually appealing pie chart with custom colors, shadows, and exploded slices.

    Args:
        scores (dict): A dictionary containing model names as keys and accuracy scores as values.
        title (str, optional): The title of the pie chart. Defaults to 'Model Accuracy Comparison'.
    """

    # Get the model names and accuracy scores
    labels = list(scores.keys())
    values = list(scores.values())

    # Create a pie chart with enhancements
    plt.figure(figsize=(8, 8))
    plt.pie(
        values,
        labels=labels,
        autopct='%1.1f%%',
        startangle=90,
        colors=colors[:len(labels)],  # Use colors based on the number of models
        shadow=True,  # Add shadow for depth
        explode=[0.05] + [0] * (len(labels) - 1),  # Explode the first slice
        textprops={'fontsize': 12, 'weight': 'bold'},  # Adjust label font
    )
    plt.title(title, fontsize=16, fontweight='bold')
    plt.show()

# Call the function to create the pie chart
create_beautiful_pie_chart(Scores_ml)
pipeline_ls = make_pipeline(CountVectorizer(tokenizer = RegexpTokenizer(r'[A-Za-z]+').tokenize,stop_words='english'), LogisticRegression())
```

```
##(r'\b(?:http|ftp)s?://\S*\w|\w+|[^\w\s]+') ([a-zA-Z]+)([0-9]+)  -- these tolenizers giving me low accuray
trainX, testX, trainY, testY = train_test_split(phish_data.URL, phish_data.Label)
pipeline_ls.fit(trainX,trainY)
pipeline_ls.score(testX,testY)
print('Training Accuracy :',pipeline_ls.score(trainX,trainY))
print('Testing Accuracy :',pipeline_ls.score(testX,testY))
con_mat = pd.DataFrame(confusion_matrix(pipeline_ls.predict(testX), testY),
          columns = ['Predicted:Bad', 'Predicted:Good'],
          index = ['Actual:Bad', 'Actual:Good'])

print('\nCLASSIFICATION REPORT\n')
print(classification_report(pipeline_ls.predict(testX), testY,
                            target_names =['Bad','Good']))

print('\nCONFUSION MATRIX')
plt.figure(figsize= (6,4))
sns.heatmap(con_mat, annot = True,fmt='d',cmap="YlGnBu")
pickle.dump(pipeline_ls,open('phishing.pkl','wb'))
loaded_model = pickle.load(open('phishing.pkl', 'rb'))
result = loaded_model.score(testX,testY)
print(result)
predict_bad = ['yeniik.com.tr/wp-admin/js/login.alibaba.com/login.jsp.php','fazan-pacir.rs/temp/libraries/ipad','tubemoviez.exe','svision-online.de/mgfi/administrator/components/com_babackup/classes/fx29id1.txt']
predict_good = ['youtube.com/','youtube.com/watch?v=qI0TQJI3vdU','uniuyo.edu.ng/eportals/','restorevisioncenters.com/html/technology.html']
loaded_model = pickle.load(open('phishing.pkl', 'rb'))
#predict_bad = vectorizers.transform(predict_bad)
# predict_good = vectorizer.transform(predict_good)
result = loaded_model.predict(predict_bad)
result2 = loaded_model.predict(predict_good)
print(result)
print("*"*30)
print(result2)
```